

PATHOLOGICAL STUDIES ON THE DRY ROOT ROT
[*Rhizoctonia bataticola* (Taub.) Butler]
OF CHICKPEA



THESIS
SUBMITTED IN
PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
IN
PLANT PATHOLOGY

2001

BY
Rakesh Kumar Prajapati

Id. No. CA-95/3709/97

DEPARTMENT OF PLANT PATHOLOGY
CHANDRA SHEKHAR AZAD UNIVERSITY OF AGRICULTURE AND TECHNOLOGY
KANPUR - 208 002 (INDIA)

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DEDICATED TO LATE GRAND MOTHER

Dr. S.S.L. Srivastava
M. Sc. (Ag.), Ph. D., F.P.S.I.
Associate Professor

Department of Plant Pathology
Chandra Shekhar Azad University
of Agriculture & Technology,
Kanpur - 208 002.



Dated : 24th Nov., 2001

CERTIFICATE

*This is to certify that the thesis entitled "**Pathological studies on the dry root rot [Rhizoctonia bataticola (Taub.) Butler] of chickpea**" submitted in partial fulfilment of the requirements for the degree of **Doctor of Philosophy in Plant Pathology** of college of Agriculture, Chandra Shekhar Azad University of Agriculture and Technology, Kanpur is a record of the bonafide research work carried out by **Shri Rakesh Kumar Prajapati**, Id. No. CA-95/3709/97, under my guidance and supervision.*

The thesis embodies the work of the candidate himself. The candidate has worked under my supervision for more than 24 months.

(S. S. L. Srivastava)

**CHANDRA SHEKHAR AZAD UNIVERSITY OF AGRICULTURE
AND TECHNOLOGY, KANPUR -208 002**



Dated : 26th Nov., 2001

CERTIFICATE

We, the undersigned members of Advisory Committee of Shri Rakesh Kumar Prajapati, Id. No. CA-95/3709/97, a candidate for the degree of Doctor of Philosophy in Plant Pathology, agree that the thesis entitled "Pathological studies on the dry root rot [Rhizoctonia bataticola (Taub.) Butler] of chickpea", may be submitted by him in partial fulfilment of the requirements for the degree.

(S. S. L. Srivastava)
Chairman
Assoc. Professor
Deptt. of Plant Pathology

(D.V. Singh)
Emeritus Scientist
Ex-Professor & Head
Deptt. of Plant Pathology

(P. S. Kendurkar)
Professor
Deptt. of Agricultural
Biochemistry

(M. Rai)
Assoc. Professor
Deptt. of Plant Pathology

(H. K. Nigam)
Assoc. Professor
Deptt. of Agricultural
Economics & Statistics

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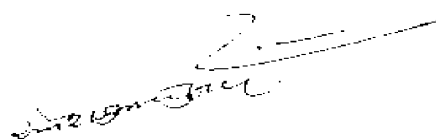
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DATE : 24th Nov., 2001



(**RAKESH KUMAR PRAJAPATI**)

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INTRODUCTION

INTRODUCTION

Chapter - 1

Pulse crops play an important role in Indian agriculture. Besides being rich in protein, they sustain the productivity of the cropping systems. Their ability to use atmospheric nitrogen through biological nitrogen fixation (BNF) is economically more sound and environmentally acceptable.

Pulses are gaining importance globally, as they are the cheap source of vegetable protein. The demand for protein is increasing by about 3 per cent annually. Awareness about the utility of pulses in human food, animal feed and soil health is increasing. However, with increase in the world population, per capita availability of pulses is getting reduced. With the increased irrigated area, pulses are further being pushed to marginal and sub-marginal lands because of their ability to survive under adverse conditions. They face tough competition with commercial crops, which have also greater stability in production.

World production of pulses now stands at 58.0 million tonnes. Chickpea (*Cicer arietinum* L.) accounts for 12 per cent (71 million tonnes) from 10.4 million ha. and traditionally grown in many parts of the world - Asia, Africa, Europe and North and South America, but the bulk of it is produced and consumed in South Asia and increasingly the Middle East and some Mediterranean countries. In these and many other regions the area of chickpea is often under reported, presumably because of the relatively insignificant position of the crop and its secondary status as a companion crop in intercropped systems (Jodha and Subba Rao, 1987).

In India, the total food production in 1999-2000 was 199 million tonnes, out of this only 15.89 million tonnes was contributed by pulses. The availability of pulses per capita 40g/day (Anon., 2000). According to an estimate and recommendation of World Health Organization the requirement of pulses for an individual is 80 g/day.

Chickpea (*Cicer arietinum* L.), commonly known as **Gram** or **Bengal Gram** or **Garbanzo bean** or **Spanish pea** has been known in this country for a long time. It is said to be one of the oldest pulses known and cultivated from ancient times both in Asia and Europe. Its probable place of origin lies in South Western Asia, that is, in the countries lying to the North West of India such as Afghanistan and Persia. According

to Aykroid and Doughty (1964) the centre of origin of chickpea is Eastern Mediterranean. According to de Candolle (1886), the fact that chickpea has a Sanskrit name would indicate that the crop has been under cultivation in India longer than in any other country.

Chickpea occupies a position of pride among the leguminous crops owing to its great importance both as vegetable and as pulse. In a country like India where most of the population is primarily vegetarian, chickpea has a special place in the daily diet of people due to its high protein contents and manifold uses. A large population in this country is already suffering from protein malnutrition to short supply and high prices of pulses, which have gone beyond the reach of poor man. Therefore, if the over increasing population is to be feed properly in our country, every effort must be made to boost up the production of this important pulse and vegetable crop.

Chickpea belongs to the sub-family Papilionatae of family Leguminosae. It has been classified in two broad groups - (1) Desi or Brown (*Cicer arietinum* L.): In this group the colour of the seed ranges from yellow to dark brown. Seed size is usually small. It is the most widely grown group. Plants are small with good branching ability. The chromosome number is $2n = 14, 16$. (2) Kabuli or white (*Cicer kabulium*): In this group the colour of the seed is usually white. Grains are bold and attractive. Yield potential of this group is poor as compared to Desi or brown chickpea. Plants are generally taller than the Desi chickpea and stand more or less erect. The chromosome number is $2n=16$.

The roots usually include a central strong tap root, with numerous lateral branches that spread out in all directions in the upper layer of soils. There are numerous nodules which fix up atmospheric nitrogen.

Among the various grain legumes dry chickpea is of world wide importance and stands first both in production and acreage of pulses in the world. In India, chickpea is grown in almost all the states both for dry grain and green pod. It is cultivated on 7.1 million hectare with annual production 5.754 million tonnes and average yield 8.1 quintal per hectare was during 1996-97. In Uttar Pradesh it is cultivated on 0.9165 million hectare with annual production 0.8524 million tonnes and average yield 9.30 quintal per hectare (Anon., 1999). Madhya Pradesh has first place in acreage and production. The other states of Northern India, growing chickpea

are Rajasthan, Hariyana, Maharashtra, Bihar, Gujarat, Punjab, West Bengal, Andhra Pradesh, Orissia and Jammu & Kashmir in sufficient area (Anon., 1986).

The chickpea is consumed in different forms. Dry chickpea are used as split chickpea (Dal) and Besan for various salty and sweets preparations. Both husks and bits of Dal are valuable cattle feed. Fresh green leaves are used as vegetable (Sag). Straw of chickpea is an excellent fodder for cattle. The grains are also used as vegetable (Chhole). It is the leading vegetable among frozen foods chickpea considered to have medicinal effect and it is used for blood purification.

Nutritionally, it is very rich in protein and a good source of essential amino-acids, minerals and vitamins (Table - 1 and 2).

Table 1 : Nutritional composition of dry chickpea seeds per 100 gm of dry edible part.

S. No.	Constituents	Content
1.	Calorific value	360 cal.
2.	Crude protein	21.1 g
3.	Fat	5.3 g.
4.	Carbohydrate	60.9 g.
5.	Minerals	
	Calcium	202.0 mg.
	Phosphorus	312.0 mg.
	Iron	10.2 mg.
6.	Vitamins	
	Vitamin 'C'	3.0 mg.
	Small amount of vitamin 'B' complex.	
7.	Fibre	3.9 g.
8.	Moisture	9.8 g.

(Singh, 1998)

Plate -I

Fig. 1: Field showing large patches of dry plants affected by *Rhizoctonia bataticola*.



Fig. 1

Table 2 : Essential and non-essential amino-acids composition of chickpea protein (g/16 gm N).

S. No.	Amino-acids	Content
	Essential	
1.	Isoleucine	4.08
2.	Leucine	7.00
3.	Lysine	6.46
4.	Total aromatic amino acids	7.37
5.	Total sulphur amino acids	2.39
6.	Threonine	2.76
7.	Tryptophan	0.68
8.	Valine	4.55
	Non-essential	
1.	Arginine	10.62
2.	Glycine	3.75
3.	Aspartic acid	9.35
4.	Serine	4.60
5.	Histidine	2.82
6.	Alanine	4.11
7.	Glutamic acid	14.85
8.	Proline	3.84

(Abrol and Chatterjee, 1980)

It is grown in *rabi* season in the poor soil. It is affected by the diseases of fungal, bacterial, viral and nematodes origin. More than 50 pathogens have been reported on this crop from different parts of the world (Nene and Reddy, 1987)

However, during the recent surveys conducted at different locations in Uttar Pradesh, the incidence of dry root rot (Fig.1) caused by *Rhizoctonia bataticola*

(Taub.). Butler was observed in moderate to severe forms. In India, it was reported by Dastur from West Bengal in 1947, but not much work has been done, so far, on this disease of chickpea. Therefore, in view of the seriousness of the disease and importance of the crop, it was thought necessary to study in detail regarding the disease as well as its management by various methods to minimize the losses in chickpea crop with the following main objectives.

OBJECTIVES:

1. Survey and collection of diseased material from different agro-climatic regions of Uttar Pradesh.
2. Symptomatology, isolation, purification of the pathogen and pathogenicity test.
3. Morphological and cultural characters of the pathogen leading to its identification.
4. Studies on the optimum growth requirements of the pathogen.
5. Production of toxin by the pathogen *in vitro*.
6. Studies on survival of the pathogen.
7. Host range of the pathogen.
8. Screening of chickpea germplasm for disease resistance.
9. Evaluation of fungicides, non-target pesticides and bio-agents against the pathogen *in vitro* and *in vivo*.

REVIEW OF LITERATURE

REVIEW OF LITERATURE

Chapter - 2

Chickpea (*Cicer arietinum*. L.) being an important pulse and vegetable crop, has attracted the attention of plant pathologists all over the world. wherever this crop is grown. A critical review of literature has revealed that chickpea suffers from more than 50 pathogens of diseases caused by fungi, in addition to bacteria, viruses and nematodes origin (Nene and Reddy, 1987) which lower the quality and quantity of the product of this crop. Some of the common fungal diseases and their causal organisms reported on this crop are given below.

Table 3 : Major fungal diseases of chickpea along with their causal organisms reported from India.

S.No.	Disease	Causal organism	Reported by
A	Affecting aerial plant parts		
1.	Floral blight	<i>Alternaria alternata</i>	Vishwakarma and Basuchaudhary (1947)
2.	Alternaria blight	<i>Alternaria circinans</i>	Gurha <i>et al.</i> , (1981)
3.	Alternaria blight	<i>Alternaria tenuissima</i>	Narain <i>et al.</i> , (1990)
4.	Ascochyta blight	<i>Ascochyta rabiei</i>	Aga (1966)
5.	Botrytis gray mold	<i>Botrytis cinerea</i>	Joshi and Singh (1969)
6.	Colletotrichum blight	<i>Colletotrichum dematium</i>	Mishra <i>et al.</i> , (1975)
7.	Phoma blight	<i>Phoma medicaginis</i>	Haware and Nene (1981)
8.	Blight of chickpea	<i>Phyllosticta rabiei</i>	Aujla (1960)
9.	Sclerotinia stem rot	<i>Sclerotinia sclerotiorum</i>	Bedi (1956)
10.	Stemphylium blight	<i>Stemphylium sarciniforme</i>	Das and Sen Gupta (1961)
11.	Rust	<i>Uromyces ciceris-arietini</i>	Asthana (1957)
B.	Affecting root/ stem		
1.	Seed and root rot	<i>Alternaria state of pleospora</i>	Haware and Nene (1976)
2.	Fusarium wilt	<i>Fusarium oxysporum</i> f. sp. <i>ciceri</i>	Mathur and Chauhan (1968)
3.	Black root rot	<i>Fusarium solani</i>	Grewal <i>et al.</i> , (1974)
4.	Foot rot	<i>Operculella padwickii</i>	Anon., (1947)
5.	Phytophthora root rot	<i>Phytophthora megasperma</i>	Vock <i>et al.</i> , (1980)
6.	Pythium root and seed rot	<i>Pythium ultimum</i>	Anon., (1954)
7.	Dry root rot	<i>Rhizoctonia bataticola</i>	Dastur (1947)
8.	Collar rot	<i>Sclerotium rolfsii</i>	Anon., (1947)
9.	Verticillium wilt	<i>Verticillium albo-atrum</i>	Erwin (1958)

Among the fungal diseases listed above on chickpea, dry root rot caused by *R. bataticola* contributed a lion's share in reducing the biomass, yield and quality of the seeds. There is no much work carried out, so far on this disease of chickpea. However, works done on chickpea (*R. bataticola*) which is reviewed here along with information available on *Macrophomina phaseolina* (Tassi.) Goid.

HISTORICAL VIEW ON TAXONOMY AND NOMENCLATURE OF THE PATHOGEN:

The fungus, which occur in asexual or imperfect state belong to the sub-division- Deuteromycotina, class- Coelomycetes, family- Sphaeropsidaceae as they form the pycnidia in which conidia (pycnidiospores) reproduced. de Candolle founded the genus *Rhizoctonia* with the type species *R. crocorum* (Pers.) DC. Halsted (1890) recognised *R. bataticola* (Taub.) Butler as a plant pathogen and described that black-rot of sweet potato (*Ipomoea batatas* L.) caused by *Sphaeronema fimbriatum* (E.& H.) Sacc. He further observed that the roots were filled with minute irregular masses appearing as bits of "Charcoal". Burnette (1894) describing black-rot of sweet potato caused by *Ceratocystis fimbriata* and found a similar fungal growth and sclerotia in the tissues.

Tassi (1900) first described the fungus from common bean as *Macrophoma phaseoli*. Shaw (1912) described a sterile fungus with black sclerotia in India, causing seedling disease on jute (*Corchorus capsularis* L.), cowpea (*Vigna unguiculata* L.), ground nut (*Arachis hypogaea* L.) and cotton (*Gossypium herbaceum* L.). He identified it as *Rhizoctonia solani* Kuhn on the basis of right-angle branching of its hyphae. Taubenhaus (1913) studied the symptoms caused by *S. fimbriatum* and the sclerotia producing fungus on *I. batatas* and concluded that two fungi were not related. He placed later in the genus *Sclerotium* because of the absence of spores and gave the species name "*bataticola*". Butler (1918) described the sclerotia of *R. bataticola* which have uniform texture without cell differentiation except that the inner cells may be lighter than outer cells. The mycelium of *R. bataticola* has characters corresponding to those of *Rhizoctonia*, right-angle branching, constriction of the branch at the point of origin and the formation of a septum in the branch near the point of origin.

Briton-Jones (1925) transferred the fungus into the genus *Rhizoctonia* based on the **Butler's (1918)** identification of culture. **Ashby (1927)** accepted *Macrophomina* and rejected the binomial *Macrophoma phaseoli* as the pycnidial stage of *R. bataticola*. **Haigh (1930)** classified the *R. bataticola* into three groups depending upon sclerotia size : Group 'A' with a sclerotia of 200 µm or larger; Group 'B' with sclerotia of 120 to 200 µm. and Group 'C' with sclerotia of 120 µm or smaller. He noted that some isolates in Group 'C' formed pycnida either in culture or on host plants, while isolates of Group 'A' and 'B' did not.

Kendrick (1933) suggested that the pycnidial name should be applied only to the pycnidium forming isolates whereas non-pycnidia isolates be referred to by the sclerotial name *R. bataticola*. Therefore, the name *R. bataticola* be used to designate the sclerotial isolates or the isolates belonging to Haigh's group 'A' and 'B' and pycnidial strains and the isolates of group 'C' should be called *M. phaseolina* (*R. bataticola*). **Henson and Valleau (1937)** objected to the use of *R. bataticola* over *S. bataticola* because the *Rhizoctonia* has been shown to have a perfect stage in the Basidiomycotina whereas *R. bataticola* with its pycnidial stage can not be considered as such, and real clamp-connections were never observed in *R. bataticola*. **Goidanich (1947)** examined the original Tassi's material and compared it with *M. phaseoli* and found all of them identical with *M. phaseolina*. **Kulkarni and Patil (1968)** found that their isolates from *Sesami* sp. were of the 'sesamica' type in mycelial growth but were not slow growing persistent and not formed concentric rings but were pathogenic to beans and produced concentric rings only at 35°C.

Parameter and Whitney (1970) described on the basis of sclerotia characteristics of the fungus and separated into two genera. The basic characters of *Rhizoctonia* are the production of sclerotia of uniform texture, similar cells and hyphal thread emanating from the sclerotia. The sclerotia of *Sclerotium* are made up of three zones; epidermis, cortex and medulla within the entire Sclerotium. **Dhingra and Sinclair (1973)** isolated *R. bataticola* from different parts of a single soybean plant and found differences in growth rate, pathogenicity and the characteristics of the mycelium. **Dhingra and Sinclair (1978)**. reported that the pycnidial stage of *R. bataticola* is *Macrophomina phaseolina* (Tassi.) Goid. It is known to cause diseases on

different cultivated, ornamental, medicinal and wild host plants in different parts of the world.

According to the International Code of Botanical Nomenclature the binomial *M. phaseolina* is the valid name for the pycnidial stage of *R. bataticola*. Occurrence and incidence of the dry root rot caused by *R. bataticola* was first reported in cluster bean, Lady's finger and brinjal by **Butler and Bisby (1931)** from Madras. The first report of *Rhizoctonia bataticola* on chickpea was made by **Dastur (1947)** from West Bengal. **Tripathi and Sharma (1983)** reported that the incidence of *R. bataticola* on chickpea was high from late October to mid-November, decreased in December and January, and increased in February and March. According to them, seed sown at a depth of 15cm developed less root rot than sown at only 7.5cm. **Gangwar (2000)** reported 19 to 36 per cent incidence of chickpea dry root rot from 10 locations of Uttar Pradesh.

Symptomatology :

Chattopadhyay and Bhattacharya (1967) observed that once the mycelium has reach the host surface, the fungus colonizes on the host surface without penetration. **Chan and Sackston (1973)** observed that the infection by *R. bataticola* occurs in seedlings through cotyledons during emergence and through small rootlets, crown and stem tissues after the seedling stage. The fungus survives in soil primarily in the form of sclerotia, therefore, must be stimulated for germination and the resulting hyphae grow in the direction of host surface, before it can penetrate. **Ilyas and Sinclair (1974)** reported that intra-cellular colonization occurs after disintegration of pectin of the middle lamellae and outer cell wall. After the colonization of epidermal and cortical cells, the hyphae colonize the vascular system and sclerotia develop in the xylem vessels which may completely plug the vessels in which they are formed.

Dhingra and Sinclair (1978) described that the disease symptoms are due to a complex of factors. However, the main symptoms on the adult plants is dehydration of leaves, thus it can be concluded that such symptoms will be produced, if the fungal affect is, primarily, on the water transport system. Pectolytic and cellulolytic enzymes are required for invasion of the host tissues, which is also accompanied by some tissues disruption and disorganization. But these enzymes will not drastically cut

water transport in *R. bataticola* infected plants. The only factor which will primarily contribute towards the interruption of water is the production of intra-xylem sclerotia, which are formed in large numbers in xylem vessels. In seedling sclerotia are not produced but toxin produced in seedling causes dehydration of seedling leaves. Rangaswami (1994) described that when the stem examined closely, dark lesions may be seen on dark at the ground level. If the plants are pulled out from the soil and examined, the basal stem and main root may show dry root rot symptoms. The tissues are weakened and break-off easily. In advanced cases the sclerotial bodies may be seen scattered on the affected tissues.

MORPHOLOGICAL AND CULTURAL CHARACTERS OF THE PATHOGEN:

Martin (1917) first studied in detail the various steps involved in the process of sclerotia formation. Uppal *et al.*, (1936) found that sclerotia initiated with division of hyphae into barrel-shaped cells which in turn gave rise to branches that may grow outwards and become branched. Vasudeva (1937) reported that the fungus may produce abundant aerial mycelium. Hildebrand *et al.*, (1945) observed the sclerotia of *R. bataticola* are black, smooth, varying from spherical to oblong to irregular shaped. The sclerotial shape of any single isolate may vary with the substrate on which the fungus is grown. In culture or on the surface of plant tissues they are globular, oval or pear-shaped, while in cortical tissues they tend to be flattened. Reichert and Hellinger (1947) reported that in all cases the hyphae of young colonies are hyaline with granular contents, which disappear with age. They also reported five types of sclerotia development. Ahmed and Ahmed (1969) divided their eight isolates into two groups based on the mycelial color, one white to gray and other certain isolates dark gray.

El-Helaly *et al.*, (1970) reported that the morphological characters of *R. bataticola* vary with the isolate and culture's age. The fungus grows rapidly in culture and produces typical dark, generally black sclerotia. A non-sclerotia-forming mutant also observed. Ilyas and Sinclair (1974) reported that pycnidial wall is black to brown to greyish with three to four layers of thin angular cells. Indrani-Ghosh *et al.*, (1990) studied the morphology and development of sclerotia and pycnidia in case of *M. phaseolina*. Singh and Nene (1990) obtained cultures of *R. bataticola* from

chickpea, pigeonpea, groundnut and sorghum plants from ICRISAT fields and made cross inoculation studies inferring that the chickpea isolate was the most aggressive. Than *et al.*, (1991) carried out colony compatibility tests using 37 isolates of *R. bataticola* from diseased chickpea plants obtained from different locations and 2 isolates from other hosts. The results indicated possible occurrence of more than one type of isolates in a field and the existence of closely related isolates from soybean and *Phaseolus vulgaris* to chickpea isolates.

OPTIMUM GROWTH REQUIREMENTS OF THE PATHOGEN:

Post (1933) studied different natural and synthetic media for inducing pycnidia formation of which acidified corn-meal agar was most useful. Livingston (1945) found six ranked on maltose and glucose as a carbon source for sorghum isolates. Goth and Ostazisk (1965) induced sporulation in 14 isolates on propylene oxide-sterilized leaf tissues incubated at 30°C. Visible light induced quick and profused sporulation though some sporulation also occurred in the dark. Knock-Davies (1965) observed that *R. bataticola* could not induce sporulation unless the cultures were irradiated with ultra-violet light for 12 to 24 hours at 30°C. Gangopadhaya *et al.*, (1974) reported that soybean isolates grew well if galactose was provided as carbon source

Vasudeva (1936) reported optimum temperatures of 25 to 30°C for some isolates with increased growth above 30°C. Uppal *et al.*, (1936) reported that two isolates did not grow at 9°C and 11°C respectively after 2 days, but slight growth of both isolates was observed after 5 days. Similarly at 40°C, 3.9 cm. and 1.9 cm. growth occurred in 7 days, respectively. They also recorded the optimum temperature between 30 and 45.5 °C for the isolates of *M. phaseolina* from sorghum. Adam and Stokes (1942) in general found optimum growth of *M. phaseolina* in culture at 30 to 37°C depending upon the isolate. Little or no growth occurs at 10°C or below and at 40°C or above. Smith (1964) and Reynolds *et al.*, (1972) showed that fluctuation in temperature with an amplitude of 2.44°C at mean temperature of 35°C had a detrimental effect on the growth rate. Jackson (1965) recorded maximum dry weight for the fungus grown at 26 to 32 °C but the isolates varied in the total amount of growth at optimum temperatures.

Uppal *et al.*, (1936) comparing the growth of two isolates found that one isolate grew at pH 12.4 while the other did not, and both grew at pH 11.4 though the growth was below the optimum of *M. phaseolina*. Livingston (1945) reported that a sugar beet isolate of *M. phaseolina* grew very well at pH range of 5.0 to 8.0, whereas, the maximum growth of a corn isolate was in a range of 5.0 to 7.0 and dropped sharply at pH 8.0. Bhargava (1965), Shanmugam and Govindaswamy (1973) obtained good growth of *M. phaseoli* of groundnut at pH 5.0. Jain (1969) showed that the five isolates of *M. phaseolina* obtained from different parts of *urd* plant responded differently to the initial pH of the medium used. Ratnoo and Bhatnagar (1991) found an isolate of *M. phaseolina* from cowpea which grew best and formed most sclerotia on potato dextrose agars medium at 30-35°C. Growth and sclerotial formation increased with pH to a max. at 7 and then declined. Tariq-Husain *et al.*, (1995) reported that colonization of *M. phaseolina* on chickpea roots was higher when plants were subjected to moisture stress compared with unstressed plants. Colonization on primary roots was greater at early stages with secondary roots infected at later stages of plant growth.

TOXIN PRODUCTION BY THE PATHOGEN *IN VITRO* :

Mathur (1968) showed that cut shoots of sunflower when placed in autoclaved or non-autoclaved culture filtrates of *M. phaseolina* produced typical foliar symptoms of the disease. The symptoms were similar to those produced on the leaves of inoculated sunflower plants. The symptoms on shoots placed in non-autoclaved filtrate appear earlier and were more typical of the disease in nature than on shoots in autoclaved filtrate.

Chan and Sackston (1969) observed that when sunflower stem or petioles were placed in cell-free culture filtrate of *M. phaseolina* necrotic spots appeared on the leaves. The pattern of spotting was identical to that of eosin dye deposition, when sunflower shoots were placed in the dye solution. The necrotic spots appeared only after the fungus had colonized the vascular system. The severity of the necrosis was related to the degree of colonization. They also reported that toxin was not host specific and not related to the virulence of the isolates. Rai and Singh (1973) showed that toxic substance produced in the culture of *M. phaseolina* caused necrosis, chlorosis and wilting of *Brassica juncea* leaves and inhibited seed germination.

Dhingra and Sinclair (1974) tried to purify the principal toxin produced in culture and the inoculated soybean seedlings. The partially purified toxin from culture produced symptoms on the cut soybean seedlings in 30 minutes at concentration of one mg/ml and 23.5 hours at concentration of 50 µg/ml. The toxin translocated through the vascular system and not metabolized by the host tissues, it could be reisolated from the cut soybean seedlings placed in the solution.

Sitansu-Pan *et al.*, (1986) studies *in vitro* production of toxin by the isolates of *M. phaseolina* and confirmed their non-specific nature. Toxin production was not linked with virulence. Fungicide adaptation altered the amount of toxin produced as indicated by the inhibition of rice radicle growth and symptoms expression on the soybean. **Dhipanwita-Bhattacharya *et al.*, (1992)** isolated Asperlin, Isoasperlin and Acetylphormalactone from culture filtrates of *M. phaseolina*. These 3 metabolites and the previously isolated Phaseolinone and Phaseolinic acid caused non-specific leaf necrosis on several plants tested. Phaseolinone had the highest-leaf spot activity and also inhibited seed germination in tests on *Vigna mungo*. **Dhipanwita-Bhattacharya (1994)** reported that the production of Phaseolinone, a phytotoxic metabolite of *M. phaseolina* in infected *Vigna mungo* seeds grown on soil, was estimated by ELISA and HPLC. The degree of inhibition of seed germination correlated well with the amount of toxin produced; 50% inhibition was observed at a toxin level of 2.1 mg/g of wet tissue. **Kuti *et al.*, (1997)** the potential for using cell free culture filtrate (CFCF) and toxin (Phaseolinone) from *M. phaseolina* and assessed that CFCF and partially purified Phaseolinone damaged *Parthenium argentatum* seedlings over a 15-days period.

SURVIVAL OF THE PATHOGEN:

Andrus (1938) reported that in *Phaseolus vulgaris* that infected areas resulting from seed-borne inoculum always developed from the cotyledons downwards. **Sackston (1969)** reported seed-borne nature of *M. phaseolina* in *Vigna unguiculata*. **Lal and Mathur (1967)** demonstrated that *M. phaseolina* infected groundnut kernel carried the fungus in both cotyledons. **Nicholson and Sinclair (1973)** found *M. phaseolina* internally seed-borne in soybean as many as 36% of the seeds in India grow under field conditions.

Norton (1953) showed that *M. phaseolina* mycelium was capable of growing in soil without a food base but it was sensitive to other organisms in soil and measured the growth of mycelium up to 20 cm in 100 days without a food base in autoclaved soil at 35°C and with a moisture content of 18.5 to 37.3%. Kovoov (1954) recorded 12 cm growth at lower moisture levels (30% MHC) with little bacterial attack of the hyphae with increase soil moisture content (50 and 80 MHC). Byadgi and Hegde (1985) reported that the saprophytic activity of *R. bataticola* in soil was max. of soil temp. of 30°C with good growth at 35°C and was max. at 15°C. Taya *et al.*, (1988) reported that incidence of *R. bataticola* on chickpea was higher on sandy than on clay soil. Low soil moisture was conducive to disease development. Collins *et al.*, (1991) studies the germination of microsclerotia of *R. bataticola* on agar slides was greatest (95 and 96%) in pasteurized soil, germination was 4% and in rhizosphere soil, it was 36%.

Singh and Nene (1990) reported that higher counts of sclerotia of *M. phaseolina* in soil in the intercropping systems of sorghum and cowpea with pigeonpea than in single cropping system.

Rangaswami (1994) described that the fungus is a facultative parasite, capable of living saprophytically on dead organic tissues, particularly on many of its natural hosts, producing sclerotial bodies. It produces the pycnidia when the atmospheric temperature is above 30°C and the pycnidiospores remain viable for over a year. It perpetuates freely and becomes virulent when the optimum predisposing conditions in the host exist. Singh (1998) described that the sclerotia of the fungus can survive free or embedded in host tissues than in free state in the soil. Dry soils prolong survival of these sclerotia. In wet soil sclerotia can not survive for more than 7-8 weeks, while the mycelium can not survive for more than 7 days. The fungus is mainly the soil dweller and spreads from plant to plant through irrigation water, tools implements and cultural operations. The sclerotia and pycnidiospores may also become air-borne and cause further spread of the pathogen.

HOST RANGE OF THE PATHOGEN :

The fungus also attracts on various hosts and was reported to cause varied type of symptoms. It causes stem rot, root rot, fruit rot, charcoal rot and wilting. Stem rot in jute (Swada, 1916), root, stem and fruit rot in *Capsicum* spp. (Martim, 1917;

Higgins, 1923), charcoal rot in black eye cowpea (Mackie, 1934), charcoal rot and wilting in cotton (Sarejanni and Cortzas, 1935), *Zea mays* (Galloway, 1935), charcoal rot in potato (Watson, 1944), *Cicer arietinum* (Dastur, 1947), Sugarbeet (Thirumalachar and Pushkarnath, 1953), *Psium sativum* (Nema *et al.*, 1960), *Citrus vulgaris* (Rao, 1964), *Luffa cylindrica* (Rao, 1964 a), *Capsicum annum* (Rao, 1964 b), *Setaria italica* (Grewal and Mahendra Pal, 1965), Cabbage seedling (Grewal and Singh, 1965), *Arachis hypogea* (Rao, 1966), *Colocasia spp.* (Grower and Prasher, 1968), *Phaseolus aureus* (Philip *et al.*, 1969), Sorghum leaves (Raut and Bhombe, 1972), *Cynamopsis tetragonalobe* (Singh and Chohan, 1973 b).

Singh *et al.*, (1990) studied the host range of *R. bataticola* isolated from chickpea plants affected with dry root rot at ICRISAT in pots and reported new hosts and 10 species were found to be new pycnidial hosts. Songa and Hillocks (1996) investigated soil inoculum in Kenya and reported that all the legume crops and weeds tested become infected after inoculation with *M. phaseolina*, *P. vulgaris*, soybean and cowpea were the most susceptible and pigeonpea, green gram and lablab were moderately susceptible. Groundnut, chickpea *Cassia spp.* and *Crotolaria spp.* were least susceptible.

SCREENING OF CHICKPEA GERMPLASM FOR DISEASE RESISTANCE :

Krishnamohan *et al.*, (1981) tested 20 chickpea forms screened under field conditions and artificially in the green house, using a 1-6 score (1 - highly resistant), BG205 and BG206, although moderately susceptible in the green house, were highly resistant under field conditions against *R. bataticola*. Singh *et al.*, (1982) reported that 3 chickpea cultivars did not to be related to their resistance to *R. bataticola*. Carbohydrate content was higher in the susceptible cultivar than in the 2 resistant ones. Singh and Mehrotra (1982) reported resistance to *R. bataticola* was shown by the cultivars BG-203, G-543 and Hare Chhole when grown in infested soil. Reddy *et al.*, (1990) tested resistant to wilt and different root rots of chickpea and found that the mortality of variety JG-62 (100%), Avarodhi and ICC-48 was (20%). Baker and Ahmed (1991) tested the resistance of 90 genotypes of chickpea in field infested with wilt, dry root rot pathogens and found that ICC 12263 was most resistant.

Von Rheenen *et al.* (1992) reported several resistant cultivars of chickpea viz., ICC 2644, 10384, 10630, 11224, 11332, ICCL 81002, 81010, ICC 1244, 12450,

compared with 70% in the untreated controls. **Ramadoss and Sivaprakasam (1987)** tested Carbendazim, Quintozene and Thiram at 250, 500 and 1000 p.p.m. and 2 granular insecticides Carbofuran and Aldicarb at 250, 500, 1000 and 2000 p.p.m. *in vitro* against *M. phaseolina* and reported that all were inhibitory to linear growth, Carbendazim (at 10 p.p.m.) and Thiram (at 500 p.p.m.) more fungicidal while Quintozene and the insecticides were fungistatic.

Taya et al., (1990) in green house tests, in different ways and found that the best control of *R. bataticola* was given by Carbendazim alone or in combination with Thiram as seed treatment, pre-sowing, soil-drench and seed treatment + drenching after sowing treated seeds of chickpea. **Kumar and Khare (1990)** investigated the antagonistic relationship of soybean with *R. bataticola* and *Sclerotium rolfsii*. It was inferred that the populations increased and decreased due to the antagonistic activity of *T. harzianum* and *Bacillus subtilis*. **Singh et al., (1992)** reported that integrated pesticide spray schedule provided better control of dry root rot (*R. bataticola*) in chickpea than any of the simple treatments. Higher yields were obtained when seed treatment with Carbendazim + Thiram was followed by 2 sprays of Monocrotophas or Indosulfan.

Peshney et al. (1992) reported that Thiram, Captan, PCNB, Mancozeb, Iprodione, Carboxin, Carbendazim, Thiobendazole (all 0.2%) and Tridemorph (0.07%) effectively controlled growth and sclerotial germination of *R. bataticola in vitro*. Carbendazim and Thiobendazole were most effective and inhibited growth at the concentration of 0.006%. In successive cultures, the fungus should increasing tolerance to Mancozeb and PCNB and to a lesser extent to Thiram, Captan, Carboxin and Iprodione. There was no reduction in sensitivity towards Carbendazim or Thiobendazole.

Kulkarni et al., (1992) listed the efficacy of herbicides (Basalin, Diuron, Cotoran, Telvar, Ansar and Lasso) against *R. bataticola in vitro* at 500-2000 p.p.m. and found that Lasso gave maximum inhibition of fungal growth at 500 and 1000 p.p.m. Ansar was also effective at 2000 p.p.m. Diuron and Basalin were the least effective chemicals. **Singh (1995)** studied the effects of *T. harzianum* and *T. viride* on cowpea against *M. phaseolina in vitro*. Both *Trichoderma* spp. inhibited growth on potato dextrose agar medium, but *T. viride* caused greater inhibition.

MATERIALS & METHODS

MATERIALS AND METHODS

Chapter - 3

SURVEY FOR PREVALENCE AND INCIDENCE OF THE DISEASE:

To find out the prevalence and incidence of the *Rhizoctonia bataticola* causing dry root rot of chickpea, an extensive survey was conducted during the year, 1998-99 and 1999-2000 in different parts of Uttar Pradesh under various agro-climatic conditions of U.P. The locations included, Students instructional farm of C.S.A. Univ. of Agric. & Tech., Kanpur, Research farm of Indian Institute of Pulses Research, Kanpur, Instructional farm of students, N.D. Univ. of Agric. & Tech., Faizabad, Crop research farm of B.H.U., Varanasi, Crop research centre of G.B. Pant Univ. of Agric. & Tech., Pantnagar (Udhamsingh Nagar) and farmers' fields in Jaspura (Banda), Mauranipur (Jhansi), Talbehat (Lalitpur), Kadaura (Jalaun), Kabrai (Mahoba), Maudaha (Hamirpur), Ajitmal (Etawah), Naini (Allahabad), Ajagen (Unnao), Bakshi Ka Talab (Lucknow), Daurala (Meerut), Rampur (Saharanpur), Munderva (Basti), Kaptanganj (Deoria) and Meerganj (Bareilly). Totals 5 fields each location were surveyed from above twenty locations during the crop season. One hundred randomly selected plants from each field were collected at podding to pod maturity stage. All the specimens were examined in the laboratory for the presence of the causal organism. All the specimens were properly pressed dried, labeled and kept for further studies and records.

Procedure for recording disease incidence:

One hundred sampled plants from all five fields of all 20 locations were examined for the dry root rot symptoms and also examined microscopically for the disease. Based on the number of infected plants from each field, per cent dry root rot was calculated for every location.

Symptomatology of the disease under natural conditions :

The characteristics disease symptoms produced in nature on root, leaves and pods, were critically observed and recorded periodically.

Isolation and purification of the pathogen:

Infected roots, showing distinct characteristic symptoms were selected for isolation of the pathogen. The selected roots were washed with fresh and sterilized water in order to remove the soil dust particles and surface contaminants. Subsequently, young diseased parts of the root were cut into small pieces along with some healthy portion with the help of sterilized blade. These cut pieces were then dipped in 0.1 per cent solution of mercuric chloride for 30 seconds and then washed thoroughly three-four times with sterilized water to remove the traces of mercuric chloride. Excess moisture was removed by putting these pieces pressed in between two folds of sterilized blotter paper under aseptic conditions in the inoculation chamber. These pieces were then transferred with the help of sterilized forceps to Petri-dishes, which were already poured with 2 per cent potato dextrose agar medium under aseptic conditions followed by incubation at $30 \pm 1^{\circ}\text{C}$. As soon as (48-72 hours of inoculation) the mycelial growth was visible around these pieces, the hyphae tips from the advancing mycelium, were transferred aseptically into the sterilized culture tubes containing 2 per cent potato dextrose agar medium.

The culture was purified by single hyphal technique. A diluted hyphal suspension was poured on the surface of plain agar in Petri-dishes aseptically and hypha were allowed to settle down on the agar surface. The amount of suspension was so adjusted as to form a very thin layer over the surface of the agar. Hyphae, which settled apart from each other, were selected under the microscope and encircled with the help of glass making pencil. They were lifted along with agar individually and transferred to Petri-dish, already containing sterilized potato dextrose agar. After the proper growth of the culture obtained from the transferred single hyphae, subculturing was done in test tubes and purity was checked up. After purification, the culture of each isolate was multiplied and maintained at 2 per cent potato dextrose agar medium slants in the refrigerator at low temperature ($8-10^{\circ}\text{C}$) for further studies. Thus, 20 isolates of the fungus were obtained from the diseased samples of different locations. The growth characters of these isolates were studied on potato dextrose agar medium.

Pathogenicity tests :

Pot experiment using soil inoculation technique was carried out in order to determine the pathogenic capability of *R. bataticola* isolated from chickpea plant

roots. Pots filled with sterilized soil (autoclaved at 1.10Kg./cm² pressure for two hours) which was previously disinfected with 5.0 per cent solution of formaline. The inoculum was prepared by growing pure culture of *R. bataticola* on sand chickpea flour medium in 250ml conical flasks. (mixture of 10g. chickpea flour with 90g. river bed sand) in 20 ml distilled water in each flask and autoclaved at 1.10 Kg./cm² pressure for 30 minutes. Each flask was then inoculated with pure culture of *R. bataticola* and incubated at room temperature (approximately 30°C) for 15 days. Inoculation of soil was done within seven days before sowing by thoroughly mixing soil with fungus culture grown on sand chickpea flour medium in pots. The culture was added at the rate of 5% of the weight of soil in the pots. Controlled pots were filled with soil, without adding inoculum. Healthy and surface sterilized seeds of the chickpea variety JG-62 were first disinfected with 2.5 per cent sodium hypochloride solution for 3 minutes and then rinsed with sterilized water dried and sown at the rate of 5 seed per plot. The pots were then incubated in glass house at 30±1°C and irrigated regularly to maintain sufficient moisture. In control pots, sterilized chickpea flour sand medium without fungal inoculum was mixed and surface sterilized seeds of JG-62 were sown at the rate of 5 seeds per pot with four replications in each treatment. The pots kept in glass house were observed critically for seedling emergence and appearance of symptoms on seedlings and adult plants upto 60 days of sowing. The pathogen was re-isolated from the internal tissues of infected plants in potato dextrose agar medium for confirmation of Koch's postulates (1883).

MORPHOLOGICAL AND CULTURAL CHARACTERS OF THE PATHOGEN LEADING TO ITS IDENTIFICATION:

The cultural and morphological characters of the pathogen were studies on eight different natural, synthetic and semi-synthetic media and their composition is given in Appendix-1.

- A. **Preparation of the media:** -The media were prepared by the standard methods as described by **Riker and Riker (1936)**. The different constituents of media were dissolved in distilled water separately and for solidification 2 per cent agar was added. In case of potato dextrose, Oat meal, chickpea root extract and carrot root extract, extracts were taken after boiling them on a water bath. All the media were sterilized in the autoclave at 1.10 Kg/cm²

pressure for 20 minutes. The glasswares were cleaned with chromic acid and distilled water was used during the entire study relating to media suitability.

- B. **Sterilization of Petri-plates and other glasswares:-**Cleaned Corning or Borosil glasswares were used. Unless otherwise mentioned all the Petri-plates were sterilized in hot air oven at 160° for 90 minutes.
- C. **Pouring of media in Petri-plates:-**Previously sterilized Petri-dishes of 9 cm diameter were used for pouring the medium. A set of four Petri-dishes were maintained for each treatment and 20 ml. sterilized, melted but cooled medium (about 50°C) was aseptically poured in each Petri-dish. The Petri-dishes were handled aseptically in a sterilized inoculation chamber using a spirit lamp flame in the chamber. The medium in the plates was allowed to solidify before inoculation with the pathogen.
- D. **Inoculum :-** Fifteen -day-old culture of the pathogen, under study, grown on 2 per cent potato dextrose agar medium was used as inoculum for all the studies. Petri-dishes, containing an equal amount of agar-medium were inoculated with 5 mm culture discs, cut with the help of a sterilized metal cork borer from the growing region of the pathogen. A single culture disc was placed in the centre of each Petri-dish aseptically by using a sterilized inoculation loop touched the surface of the medium. Flasks, containing liquid medium, were also inoculated aseptically with 5mm culture discs of the pathogen. The inoculated Petri-dishes and flasks were incubated for ten days at temperature of $30\pm 1^{\circ}\text{C}$.

Determination of the basal medium :

For selection of the basal medium, the pathogen was grown on eight different liquid media. The average mycelial dry weight and sporulation of the pathogen were considered as the main criteria for selecting the medium. The experiment was replicated four times.

Linear growth rate:

To determine the linear growth of the pathogen, 20 ml of sterilized medium was poured in sterilized Petri-dishes. Each Petri-dish was inoculated with the equal quantity of the inoculum. These were then incubated at $30 \pm 1^{\circ}\text{C}$ for ten days there

after, the linear growth was recorded in mm two directions at right-angles to each other and the average was calculated. Cultural characters such as colony type and colour were recorded. The experiment was replicated four times.

Determination of fungal dry weight :

To determine the quantity of mycelial mat produced by the pathogen, 50 ml of each liquid medium was poured into 150 ml conical flasks. The flasks were sealed with cotton plugs tightly. These were then sterilized in autoclave at 1.1Kg/cm^2 pressure for 20 minutes. After cooling the flasks were inoculated with an equal amount of fungal inoculum and incubated at $30\pm 1^\circ\text{C}$ for ten days. After the expiry of the incubation period, the mycelial mat was filtered through Whatman's filter paper No.42, washed thoroughly with water to remove the traces of chemical associated with the mycelial mat, oven dried at 60°C for 48 hours, cooled in desicator and accurately weighted on electric balance. Three flasks, representing three replications, were kept for each medium. Average dry weight of the three replication was taken as standard value for comparing the growth under different treatments.

Sporulation :

The degree of sporulation was graded in the five categories on the basis of spore count under the low power of microscope. For this purpose, one flask of each medium was randomly selected and thoroughly shaken. From each flasks, 1 ml of fungal suspension was taken and diluted to five times with distilled water 1 ml of this diluted suspension was divided into five equal drops and placed in Haemocytometer for counting the spores under low power of microscope their numbers were the counted in each drop. Finally, the average number of the spores was calculated by dividing the total number by five and the categorization was made on the basis of the following scale.

Number of the spores	Intensity of the sporulation	Symbols
0	Nil (no spores)	-
1-10	Poor	+
11-20	Fair	++
21-30	Good	+++
30 and above	Excellent	++++

Colour determination:

Ridgways (1990) "Colour Standard and Colour Nomenclature" was used for determining the colour production by the pathogen in various basal media.

Measurement of different fungal structures:

Slides of the fungal structures including hyphae, conidia, sclerotia, pycnidia, and pycnidiospores were prepared from affected tissues and 15-day-old cultures. Temporary slides were prepared in lactophenol, containing cotton blue and then examined under a compound microscope. Size, shape, septation and colour of the sclerotia, pycnidia, pycnidiospores and hyphae were recorded. One hundred counts formed the uniform basis and the data were collected accordingly.

Identification of the pathogen:

The pathogen was identified on the basis of morphological characters as stated below.

- (a) **Colony and growth habit characters:** - The growth characters of the fungus as it appears and look on the culture which was grown on Petri-dish were recorded under stereoscopic binocular microscope (6.4 - 40x magnification). The cultural and morphological characters of the fungus were recorded in potato dextrose agar after seven days of inoculation. Colour and type of growth of the mycelium were observed microscopically.
- (b) **Mycelial characteristics:** - Septation, branching pattern colour and width of the hyphae etc. were studied under microscope for specific identification.
- (c) **Pycnidium and Pycnidiospores characteristics:** Colour, shape, size and septation of the pycnidium and pycnidiospores were recorded under the compound microscope. The size of the spores were based on measurements of 50 spores colour according to standard nomenclature for colour (Ridgway, 1990).
- (d) **Sclerotia:** The shape, size, colours, texture and the pattern of sclerotia of *R. bataticola* were recorded and the size was determined by measuring 100 sclerotia.

Effect of temperatures on the growth and sporulation of the pathogen:

To study the effect of temperatures on the growth and sporulation of the pathogen, it was grown at ten different temperatures viz; 15, 20, 25, 30, 32, 33, 35, 38, 40 and 45°C in different incubators. In this study potato dextrose agar was used as the basal medium and the methods of sterilization, incubation, filtration and determination of the dry mycelial weight were followed as described earlier. The sporulation and the dry mycelial weight were determined after 10 days of incubation. Three replications were kept for each treatment.

Effect of H-ion concentration on the growth and sporulation of the pathogen:

In order to study the effect of H-ion concentration on the growth and sporulation of the pathogen the pH in the basal potato dextrose agar was adjusted to twelve different levels, viz., 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0 and 9.0 before autoclaving by Beckman's pH meter using N/10 sodium hydroxide and N/10 hydrochloride acid. The method of sterilization of the medium, inoculation, filtration and determination of dry mycelial weight of the fungus were the same as described earlier. The average dry weight of the mycelium and extent of sporulation were recorded after 10 days of incubation at $30 \pm 1^\circ\text{C}$.

PRODUCTION OF TOXIN BY THE PATHOGEN *IN VITRO*:

Preparation of the culture filtrate:

The pathogen was grown in chickpea root extract medium in 150 ml Erlenmeyer flasks at $30 \pm 1^\circ\text{C}$ for 15 days. The mycelial mat was then filtered through Whatman's filter paper No. 42 and the filtrate was made free from any living organism by filtering through Seitz filter using 3 GSF filter. This filtrate served as crude toxin.

Effect of toxin on leaf:

The toxic effect of the filtrate on intact chickpea leaves was studied by the technique of Braun (1955). For this test, the leaf injury was caused with the help of sterilized needle and 2-3 drops of culture filtrate were placed at the site of injury. Development of necrotic zones around pricked areas, if any were noted after 24 hours.

Effect of toxin on seed germination and seedling vigour:

The effect of various dilutions of crude toxin, viz; 1:10, 1:50, 1:100, 1:500 and 1:1000 on seed germination was studied by soaking the surface sterilized (2.5 sodium hypochloride for 1 minute) seeds of *deshi* chickpea, for 24 hours in the filtrate of various dilutions mentioned in such a way that in each flask the filtrate remained slightly above the seeds. The seeds were then plated in 15 cm Petri-dishes with filter paper soaked in sterilized water at the rate of 50 seeds per plate. The final observation of seed germination was taken after seven days. Ten randomly selected seedlings from each Petri-plate were used for recording root and shoot length and their average was calculated for each replication. Each treatment was replicated three times. The seeds soaked separately in sterilized medium and water and germinated similarly served as control.

To study the effect of heat on the inhibitory property of the filtrate, it was autoclaved at 1.10 Kg/cm² pressure for 20 minutes before soaking the seeds in it for 24 hours and carrying out the germination test.

SURVIVAL OF THE PATHOGEN:

In order to study the survival of the pathogen, the following experiments were conducted under laboratory and field conditions.

1. Role of seed:

In order to ascertain the role of seeds as carrier of the pathogen, surface sterilized healthy seeds mixed with mycelial suspension of the pathogen and the seeds obtained from the diseased plants, were stored in bamboo paper envelopes at room temperature (15-38°C). Monthly isolations were taken from both kinds of seeds to ascertain the survival of the pathogen in association with the seeds.

In order to study the role of seeds as primary source of infection of the disease, the seeds collected from naturally infected plants as well as those mixed with mycelial suspension of the pathogen were sown in pots filled with sterilized soil. Ten seeds were sown in each pot replicated three times. Surface sterilized seeds, sown in pots having sterilized soil, served as control. The experiment was conducted in glass house to check the aerial infection and the observations on disease appearance were taken regularly.

2. Role of soil:

With a view to ascertain the role of soil in the survival of the pathogen, the fungal culture was mixed with autoclaved soil and stored at room temperature (15-38°C) in polythene bags. Monthly isolations were taken from this soil to determine the survival of the pathogen. For determining the role of infested soil as a primary source of infection, three pots were filled with the infested soil stored at room temperature. Ten surfaces sterilized healthy seeds of chickpea, obtained from healthy seeds of chickpea, obtained from healthy plants, and were sown in each pot equidistant. The pots, filled with the autoclaved soil and sown with surface sterilized healthy seeds, served as control. These pots were kept in the glass house to avoid aerial infection. The growing plants were regularly examined for disease development.

3. Role of diseased plant debris:

- (a) Under laboratory conditions: Diseased plant roots from naturally infected chickpea plants were collected in *rabi* 1999, wrapped in tissue paper after drying between blotting papers and stored in the laboratory till the next crop season. Monthly isolations were taken from this material till the start of the next season to know the viability of the pathogen.
- (b) Under field conditions in pots : Diseased plant debris, wrapped in tissue paper, was buried in the pot soil at a depth of 8 cm and kept in the open. Regular isolations at monthly intervals were taken to find out the survival of the pathogen.

In order to study the role of diseased plant debris as a primary source of infection, two sets of experiments were conducted. In one set of pots, diseased plant debris, stored at room temperature (15-38°C), was mixed with sterilized soil and filled in four earthen pots of 30-cm diameter. In another set, the diseased plant debris, which was buried in the pot soil and kept in the open at 10-49°C, was thoroughly mixed with the sterilized soil and filled in pots in three replications. All these pots were sown with the surface sterilized healthy seeds at the rate of 5 seeds per pots equidistant. The pots filled with sterilized soil and sown with surface sterilized healthy seeds served as control. The experiment was conducted under the glass house conditions to avoid aerial contamination. The plants were regularly observed for disease development.

HOST RANGE OF THE PATHOGEN:

The investigation on the host range of the pathogen was carried out by artificial inoculation of the plants of different families. As described for the pathogenicity test. Final data on disease incidence were recorded 15 days after inoculation. Suitable control was also maintained in each case for comparison.

SCREENING OF CHICKPEA GERMPLASM FOR DISEASE RESISTANCE:

(1) **Blotter paper technique (Nene *et al.*, 1981):** Surface sterilized (5 min. in 2.5 per cent sodium hypochlorite) seeds of the test and check lines will be sown in autoclaved river bed sand placed in 15 cm. pots separately one pot for each line. The seedlings shall be watered till these are 5 days old from sowing. Inoculum of *R. bataticola* will be multiplied in potato dextrose broth and incubated for 5 days at 25°C. The mycelial mats from these flasks shall be removed at the end of the incubation period. The mycelial mats will be added to 100 ml. sterilized distilled water and macerated in a Waring blender for 1 min. (operate the blender intermittently). This inoculum will be placed in a beaker. Uprouted 5 days old seedlings of the test lines shall be washed in running water and rinse in sterilized distilled water. The roots of all seedlings of a test lines will be dipped in the inoculum for about 30 seconds. Excess inoculum shall be removed by touching the edge of the beaker. Twenty seedlings of each test line shall be placed side by side on a blotter paper (size 45 cm x 25 cm width one fold) so that only the cotyledons and roots are covered and the green top of the seedlings remain outside the blotter paper after it is folded. Fold the blotter paper and moisten it. One folded blotter paper will contain seedlings of one test line. Inoculated seedlings of a susceptible check (e.g. JG-62) with each batch of test seedlings will also be maintained and stored in a tray. These trays will be incubated in a incubator at 35°C for 8 days having 12 hour cycles of artificial light. The blotters will be kept moist by adding sterilized water everyday. At the end of the incubation period (8 days) the seedlings will be examined for the extent of root infection and scored for the disease using following scale.

Rating	Disease intensity	Reaction
1	No infection on roots	Resistant
3	Very few small lesions on roots	Moderately resistant
5	Lesions on roots clear but small; new roots free from infection	Tolerant
7	Lesions on root many; new roots generally free from lesions.	Moderately susceptible
9	Roots infected and completely discoloured	Susceptible

(ii) **Testing under artificial conditions:** The varieties found, resistant and moderately resistant under natural conditions were subjected to artificial testing in epiphytic conditions of the disease. These tests were carried out in two crop seasons. in order to locate the source of resistance. For this test, the methods adopted were the same as described for pathogenicity test. Observations on the incidence of disease were started after 20 days of sowing and continued upto maturity.

MANAGEMENT OF THE DISEASE:

For the management of this disease, the well-recognized methods of seed treatment of fungitoxicans were adopted in the present investigation. However, prior to under taking these studies, the efficacy of various fungitoxicans belonging to different groups, was first ascertained in bioassay test against the pathogen in the laboratory and those found effective were used for further experiments in pots for management of the disease. The procedure adopted in these studies were as follows:

Laboratory bioassay of different fungitoxicans: The relative efficacy of twenty one fungitoxicans, at the recommended doses, was tested against the pathogen in the laboratory (Appendix- II) Poison food technique devised by Schmitz (1930) was followed for bioassay of fungitoxicans. The requisite quantities of the fungitoxicans were incorporated into 100 ml of 2 per cent sterilized unsolidified potato dextrose agar and shaken well to make it homogenous. Medium was then poured in 9 cm. sterilized Petri-dishes with three replications of each treatment and allowed to solidify. These dishes were then inoculated from 15-day-old culture and one disc was

placed in the center of each Petri-dish in such a way that the fungus may come in direct contact with the medium. The medium, without any fungitoxicant, poured and inoculated similarly, served as control. The Petri-dishes were incubated at $30 \pm 1^\circ\text{C}$ for 15 days. The efficacy of various fungitoxicants was assessed by measuring the radial growth of the fungal colony in mm. The fungitoxicants, which were found effective or partially effective in laboratory evaluations, were further tried as seed dressers fungitoxicants for the management of the disease. .

Management of disease by seed treatment: To test of fungitoxicants, found effective in bioassay test, as seed dresser, in improving seed germination and controlling seedling infection of *R. bataticola* in chickpea, the naturally infected seeds of highly susceptible variety JG-62 were treated with them by mixing their required quantities and shaking the seeds in plugged conical flask for 15 minutes for proper mixing of the fungitoxicants with the seed. The treated seeds were sown in 30-cm diameter polythene bags equidistant @ 5 seeds per pot, replicated four times. Pots sown with untreated seed served as control. subsequently, the bags were transferred to the glasshouse benches and data on seed germination and seedling infection were recorded ten and fifteen days after sowing, respectively.

STATISTICAL ANALYSIS:

All the data analyzed statistically and statistical significance at 5 per cent level was the basis of interpreting the result. The data of disease incidence were transformed into angle

$$(\phi) = \sin^{-1} \sqrt{\text{percentage and then analyzed}}$$

EXPERIMENTAL FINDINGS

EXPERIMENTAL FINDINGS

Chapter-4

PREVALENCE INCIDENCE OF THE DISEASE:

No information is available, so far, on the extent of prevalence and incidence of the dry root of chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler in Uttar Pradesh. Therefore, with a view to assess the prevalence and incidence of this disease, a survey was conducted in different agro-climatic regions of this state during late-October to March, in 1998-99 and 1999-2000. Observations were recorded on institutional farm and farmers' field under natural conditions. Data were recorded on crops at different locations and the results of survey are presented in (Table 4 and Histogram 1)

Table 4: Incidence of the dry root rot of chickpea caused by *R. bataticola* at different locations in Uttar Pradesh.

S.No.	Location	% dry root rot
1.	Jaspura (Banda)	42
2.	Mauranipur (Jhansi)	41
3.	Talbehat (Lalitpur)	40
4.	Kadaura (Jalaun)	38
5.	Kabrai (Mahoba)	37
6.	Maudaha(Hamirpur)	36
7.	Ajitmal (Etawah)	34
8.	S.I.F., C.S.Azad Univ. of Agric. & Tech. (Kanpur)	32
9.	R.F., I.I.P.R.(Kanpur)	32
10.	Naini (Allahabad)	30
11.	Ajagen (Unnao)	29
12.	Bakshi Ka Talab (Lucknow)	28
13.	C.R.F., B.H.U.(Varanasi)	26
14.	Daurala (Meerut)	23
15.	Rampur (Saharanpur)	22
16.	Munderva (Basti)	21
17.	Kaptanganj (Deoria)	20
18.	I.F.S., N.D. Univ. of Agric. & Tech. (Faizabad)	20
19.	C.R.C., G.B. Pant Univ. of Agric. & Tech., Pantnagar (Udhamsingh Nagar)	19
20.	Meerganj (Bareilly)	19
Average		29.45

Histogram-1 : Prevalence and incidence of the dry root rot of chickpea caused by *R. bataticola* at different locations in Uttar Pradesh

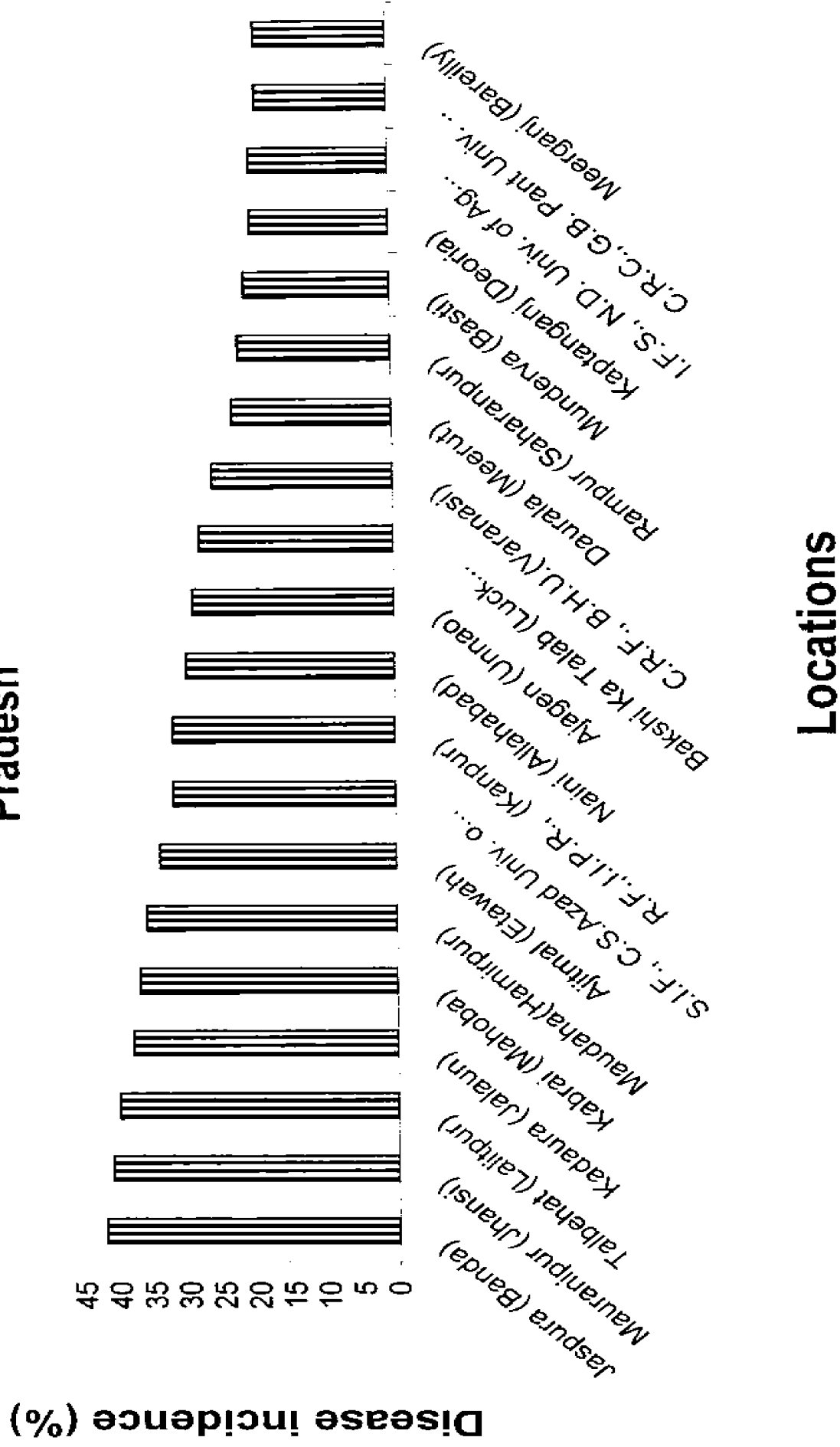


Plate -II

Fig. 2: Chickpea plant showing dry root rot symptom.

Fig. 3: Symptoms of dry root rot. Note the lack of finer roots.

Fig. 4.: Showing the sclerotial bodies on the internal part of the affected tissues of the chickpea root.



Fig. 2



Fig. 3



Fig. 4

The results presented in table reveal that the disease was prevalent in all the areas surveyed showing its wide spread occurrence. The disease incidence in different localities varied from 19.00 to 42.00 per cent. The maximum disease incidence was recorded at the Jaspura (Banda), followed by Mauranipur (Jhansi), Talbehat (Lalitpur), Kadaura (Jalaun), Kabrai (Mahoba), Maudaha (Hamirpur), Ajitmal (Etawah) and S.I.F., C.S. Azad Univ. of Agric. & Tech., Kanpur. The Lowest disease incidence, amongst the location surveyed, was recorded at C.R.C., G.B. Pant Univ. of Agric. & Tech. Pantnagar, (Udham Singh Nagar) which was similar to Meerganj, (Bareilly) .

As regards the incidence of disease in different type of variety, the maximum disease was observed in bold seeded varieties followed by medium bold and small seeded varieties.

SYMPTOMATOLOGY OF THE DISEASE:

The incidence of dry root rot of chickpea caused by *R. bataticola* was high in late Oct. to Mid Nov. decreased in Dec. and Jan., but it again increased in Feb. and March when temperature increased. The symptoms of the disease (Fig. 2 & 3) were yellowing of the leaves, within a day or two such leaves drop and plant showed wilting symptom within a week after the appearance of first symptom. When roots were examined closely dark lesions were seen. If the plants were pulled out from the soil and examined, the basal stem and main root may show dry root rot symptom. The tissues were weakened and break off easily. In advance cases the sclerotial bodies may be scattered in the internal parts of the affected tissues (Fig.4).

The fungus was considered as a moderate to high temperature pathogen. The incidence of the disease was increase with rise in temperature between 28-35°C, because the disease always appeared at high temperature. High temperature coupled with low soil moisture increased the disease incidence in double. The disease was more severe in sandy soils than heavy soils.

ISOLATION AND PURIFICATION OF THE PATHOGEN:

Isolations were made from diseased root, obtained from different locations of Uttar Pradesh. The pathogen was readily isolated on 2 per cent potato dextrose agar medium by the usual technique. Each culture was later on purified by the single hyphal culture technique described under Materials and Methods”.

PATHOGENICITY TEST:

Twelve isolates of *R. bataticola* from different samples were tested for their pathogenicability as described under "Material and Method". The results (Table 5 and Histogram 2) showed adverse effect of *R. bataticola* on the seed germination. The germination of chickpea seeds ranged between 80-90 per cent (mean 83 per cent) as compared to 90 per cent in control.

The dry root rot percentage recorded after two months of sowing, showed 2.5-50 per cent incidence in pots inoculated with *R. bataticola* isolates. There was no disease in uninoculated pots by test pathogen (control). After two and half months of inoculation pycnidial bodies developed on roots and stem, which later developed into cankerous growth. Pycnidial bodies were also seen beneath the bark of the stem.

Table 5: Pathogenic effects of *R. bataticola* isolates on chickpea variety JG-62.

Isolate No.	Germination (%)	Seedling height (cm)	Mortality (%)
1.	80	30.00	50.00
2	80	32.50	37.50
3.	90	29.00	37.00
4.	80	28.50	36.50
5.	80	30.30	37.50
6.	80	29.50	50.00
7	80	30.20	50.00
8.	80	27.50	54.00
9.	86.	32.00	38.00
10.	80	31.70	50.00
11.	90	35.60	32.50
12	90	28.40	55.00
Control	90	52.5	0.00
Average	83	30.43	44.00

Histogram-2 : Pathogenic effects of *R. bataticola* isolates on chickpea variety JG-62.

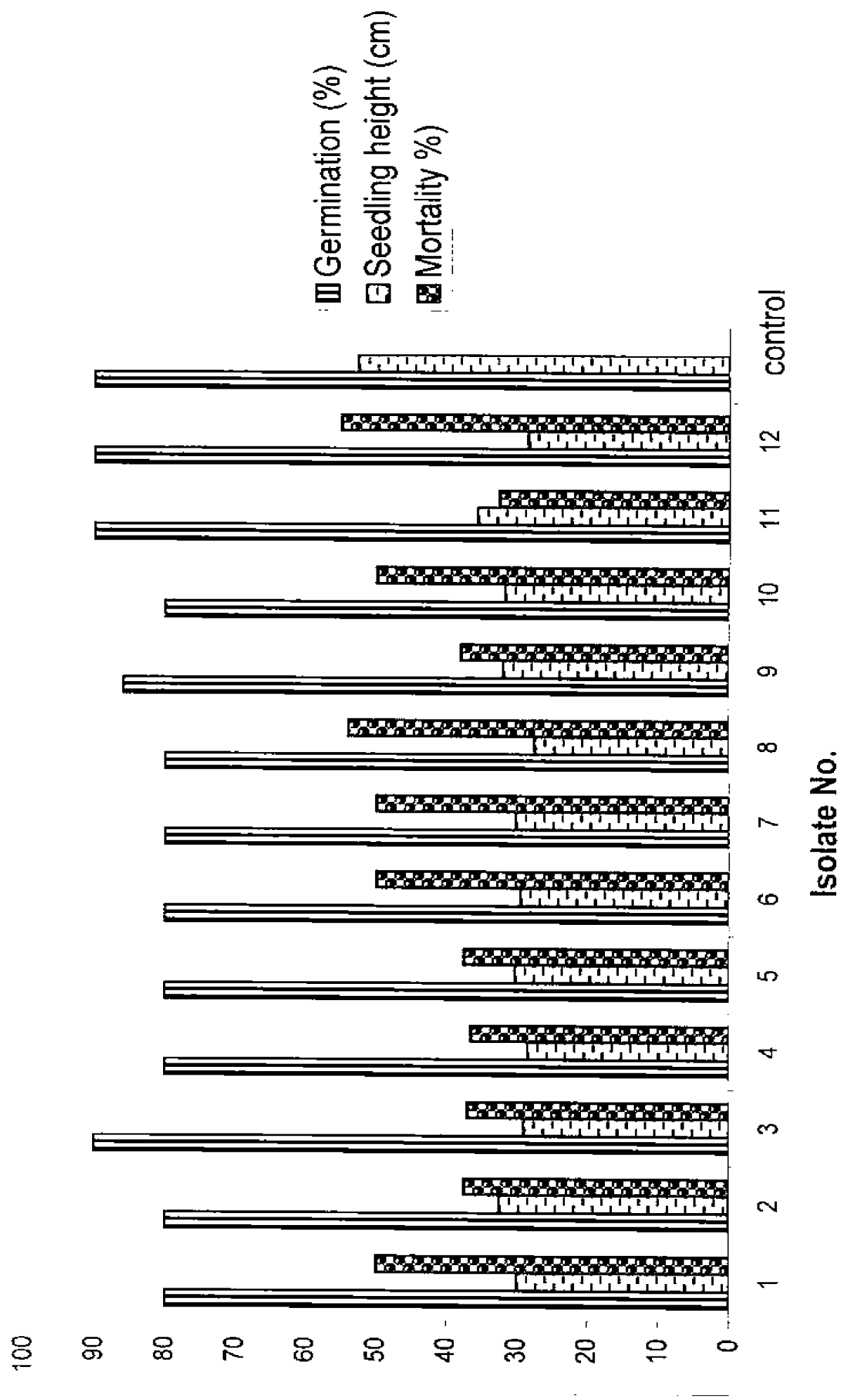


Plate -III

Fig. 5: Showing dry-up and ultimately death of variety JG-62 plants in pathogenicity test.

Fig. 6: Showing no dry root symptom of control in pathogenicity test (variety JG-62).



Fig. 5



Fig. 6

Plate -IV

Fig. 7: Showing white fluffy mycelium colonies on PDA at early stage growth of pathogen.

Fig. 8: Showing dark gray coloured colonies with fluffy mycelium on PDA later stage growth of pathogen.



Fig. 7



Fig. 8

Infected plants suddenly and prematurely dried and when such plants were uprooted their roots showed symptoms of dry root rot. The finer roots were mainly affected and had dark, blackened streaks beneath bark with dark sclerotial bodies. Such roots were brittle and broken when touched. Rotting of roots extended to the base of the stem of the plant. Early symptoms on stem and branches were spindle shaped lesions with light gray center and brown margins having scattered pycnidial bodies. The lesions coalesced and caused whole plant to dry and ultimately died (Fig.5). The fungus was reisolated from the diseased tissues and compared with the original one for Koch's postulates.

MORPHOLOGICAL AND CULTURAL CHARACTERS OF THE PATHOGEN LEADING TO ITS IDENTIFICATION:

In order to identify the isolated pathogen, the morphological and cultural characters were observed by the naked eyes and with the help of microscope on potato dextrose agar medium. The measurement of the different structures were also made and are described as below:

- (a) **Growth habit characters:** Light to dark gray coloured colonies with fluffy mycelium developed on potato dextrose agar medium. The fungus produced profuse surface mycelium with sclerotial bodies. Pycnidia were formed on host tissues and were larger than the sclerotia scattered throughout the surface of the Petri-dish.
- (b) **Colony characters:** Colonies were fluffy, fast growing, dark brown with ample sclerotial production, margin smooth, covering full plate within 6-10 days. The reverse of the colony appeared light gray in colour.
- (c) **Mycelium:** Mycelium initially grayish white, became brown to dark brown with age, septated, profusely branched, 4.5-7.5 μm in width in the xylem of the root which was 3.5-7.0 μm on agar medium, septa were more abundant in older portion than the younger one. Branches arise often at right angle with constriction at their point of origin (Fig.9). Invariably there was presence of septum close to the point of origin of the branches.

Plate -V

Fig. 9: Showing typical hyphae of *Rhizoctonia bataticola*.

Fig. 10: Showing sclerotial development on PDA.

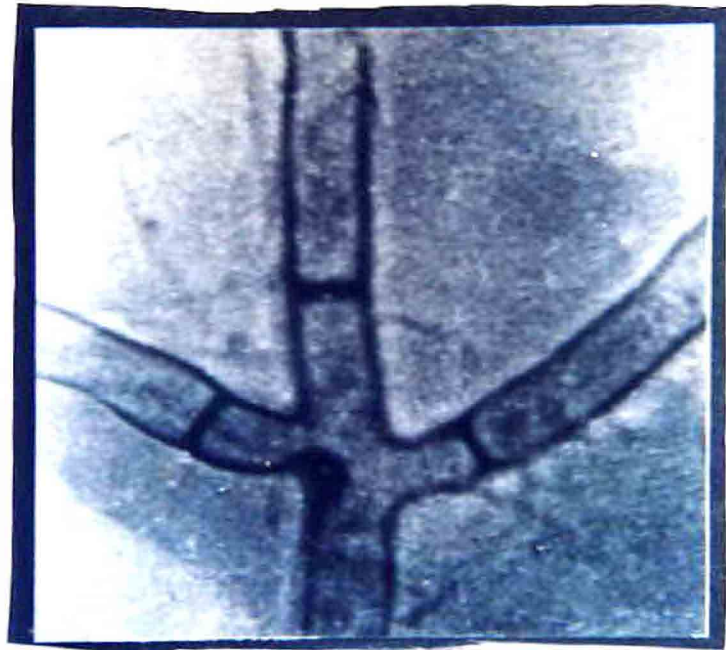


Fig. 9

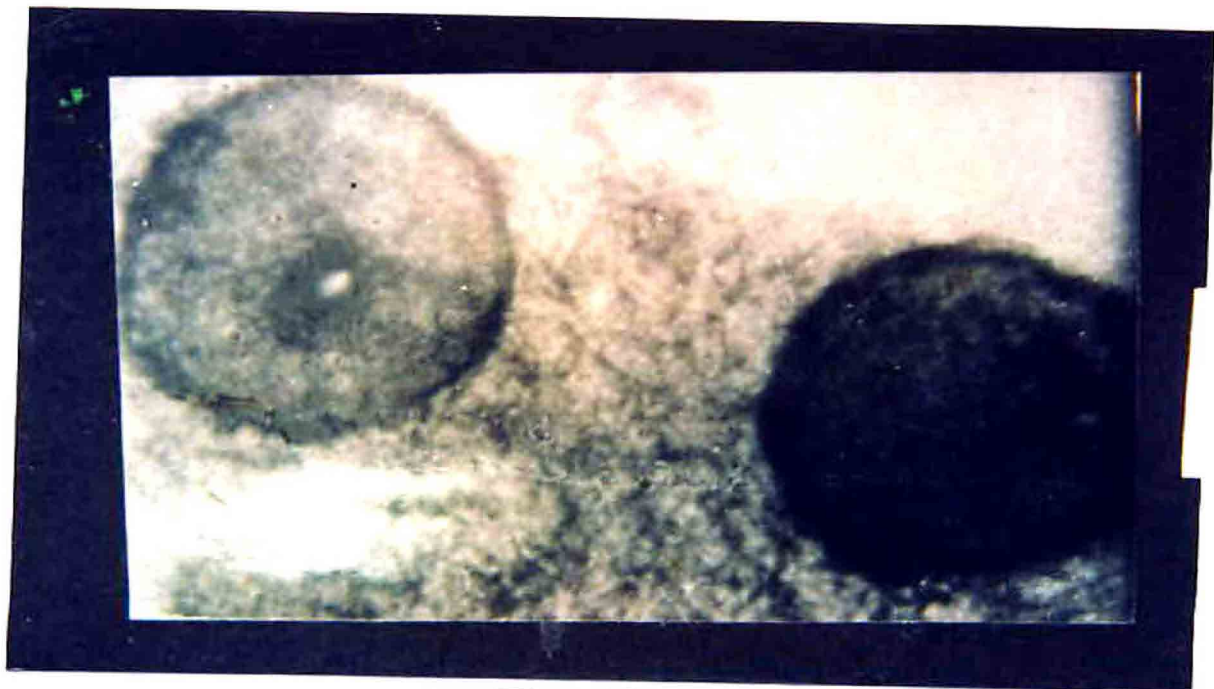


Fig. 10

- (d) **Sclerotia:** Sclerotia produced in abundance both on xylem of the root and potato dextrose agar medium were globose to oval or irregular in shape and dark brown in colour, appeared some what hairy owing to emergence of dark hyphae from their surface, measured 70.5- 95.0 μm in diameter on the xylem of the roots and always were smaller than the pycnidia produced in host tissues. (Fig .10)
- (e) **Pycnidia and pycnidiospore :** Pycnidia developed only on host tissue in field. These were globose to oval, ostiolate, beaked or without beak, light to dark brown in colour, superficial or submersed, scattered all over the outer surface of the root, measured 80.5 to 110.5 μm in diameter (Fig 11). Pycnidiospores were single celled, hyaline, oval, elongated to elliptical in shape, gazing out in the form of dull white cirrus and measured 15.5- 21 x 5.5 - 9.0 μm (Fig 12).

STUDIES ON THE OPTIMUM GROWTH REQUIREMENTS OF THE PATHOGEN :

Studies on the optimum growth requirements are important for finding out the optimum conditions for the growth of the fungus in culture. Studies were, therefore, carried out on the effect of eight natural, synthetic and semi-synthetic media, different temperatures and pH levels on the growth and sporulation of the pathogen, which are described here separately.

(A) Effect of different solid media on the growth and sporulation of *R. bataticola* :

The pathogen was grown on eight different natural, synthetic and semi-synthetic media for comparative study to select out the best medium for its growth. The data on radial growth and sporulation were recorded and the results are presented in Table- 6 .

Table 6 : Effect of various solid media on growth and sporulation of *R. bataticola*.

S. No.	Medium	Average diameter of the colony (mm)	Sporulation
1.	Potato dextrose agar	63.35	-
2.	Standard nutrient agar	51.86	-
3.	Chickpea root extract agar	47.71	-
4.	Carrot root extract agar	38.23	-
5.	Oat meal agar	29.55	-
6.	Czapek's Dox agar	25.43	-
7.	Richard's agar	17.13	-
8.	Asthana and Hawker's agar	15.26	-
	CD. at 5%	2.43	-

(-) Indicates no sporulation

A perusal of (Table-6 and Histogram-3) reveals that potato dextrose agar medium supported the best growth of the pathogen, which was significantly superior to the other media tested. Good growth of the pathogen was recorded on standard nutrient agar and chickpea root extract agar media, which did not significantly differ from each other. Fair growth of the pathogen was obtained on Asthana and Hawker's media, whereas remaining media viz, carrot root extract agar, oat meal agar, Czapek's Dox agar, and Richard's agar media supported poor growth. No sporulation was observed on any of the solid media tested.

(b) Effect of various liquid media on the growth and sporulation of *R. bataticola*.

The pathogen was grown in eight different liquid media in order to find out the best medium for its growth and sporulation. Observations on mycelial dry weight were recorded 10 days after inoculation and the results are summarized in Table-7 and Histogram-4.

Table 7 : Effect of various liquid media on the growth and sporulation of *R. bataticola*.

S. No.	Medium	Average mycelial dry weight (mg)	Sporulation
1.	Potato dextrose	22.35	-
2.	Standard nutrient	16.57	-
3.	Chickpea root extract	15.90	-
4.	Carrot root extract	14.21	-
5.	Oat meal	13.03	-
6.	Czapek Dox	12.70	-
7.	Richard's	11.55	-
8.	Asthana and Hawker's	10.12	-
	CD.at 5%	1.00	-

(-) indicates no sporulation.

It is evident from the results (Table-7 & Histogram-4), that chickpea root extract supported best growth of the pathogen and was significantly superior to the rest

Plate -VI

Fig. 11: Showing pycnidium under compound microscope developed on roots

Fig. 12: Showing single celled, hyaline pycnidiospores under compound microscope



Fig. 11



Fig. 12

of the media tested. Good growth was recorded on carrot root extract and potato dextrose medium, which were statistically at par with each other. Fair growth was obtained on oat meal medium, whereas, the remaining media namely Asthana and Hawker's, Richard's Standard nutrient and Czapek's Dox medium supported poor growth of the pathogen. Sporulation was not observed on any of the liquid media tested.

(c) Effect of different temperature on the growth and sporulation of the pathogen:

Temperature is known to have profound effect on the growth of micro-organisms. Present studies were, therefore, taken up to find out the optimum, minimum and maximum temperature requirements for the growth and sporulation of the pathogen. For this investigation, the fungus was grown at ten different temperatures ranging from 15°C to 45°C, and the average dry mycelial weights were recorded, and the results are summarized in Table-8 and Histogram-5.

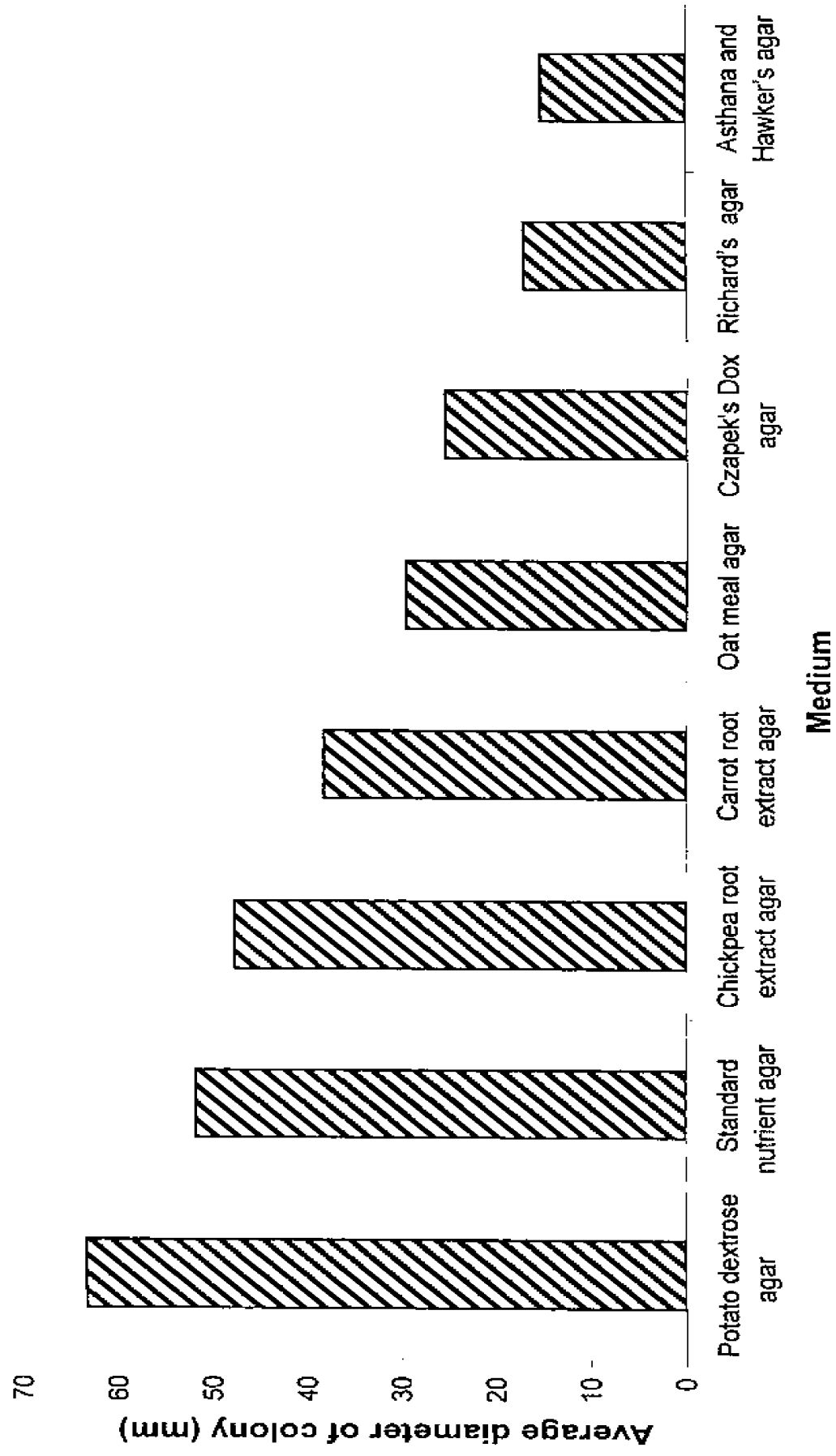
Table 8 : Effect of various temperatures on the growth and sporulation of *R. bataticola*.

S. No.	Temperature (°C)	Average mycelial dry weight (mg)	Sporulation
1.	15	6.62	-
2.	20	7.46	-
3.	25	9.30	-
4.	30	12.63	-
5.	32	15.43	-
6.	33	20.57	-
7.	35	20.70	-
8.	38	19.50	-
9.	40	19.27	-
10.	45	19.00	-
CD. at 5%		1.88	-

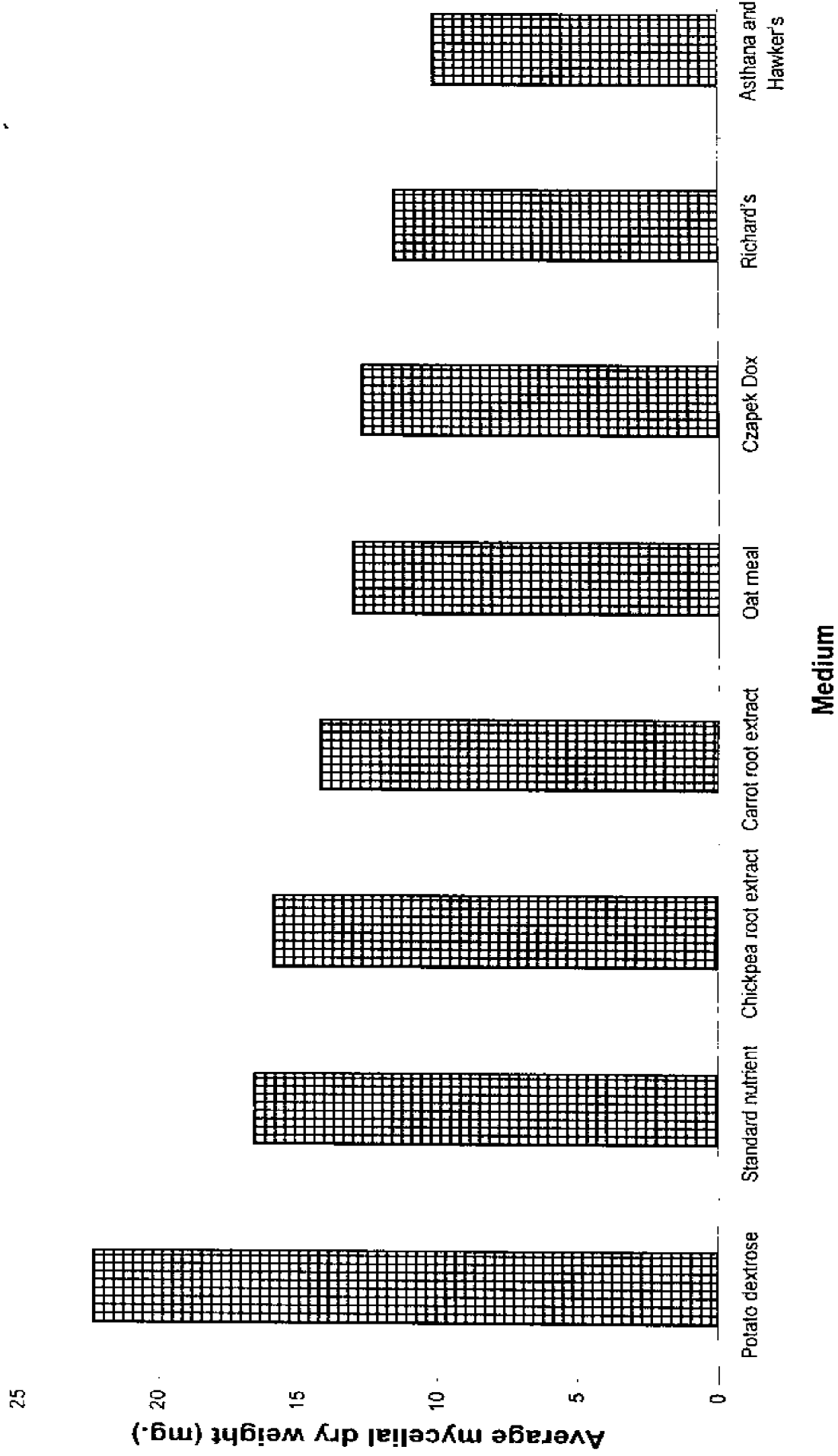
(-) Indicates no sporulation

The results (Table-8 and Histogram-5) indicate that, although, the pathogen could grow over a wide temperature range of 15°C to 45°C, but the optimum temperature for its growth was found to be 35°C. The next best temperature for its

Histogram-3 : Effect of various solid media on growth of *R. bataticola*



Histogram-4 : Effect of various liquid media on growth mycelial dry matter of *R. bataticola*



of the media tested. Good growth was recorded on carrot root extract and potato dextrose medium, which were statistically at par with each other. Fair growth was obtained on oat meal medium, whereas, the remaining media namely Asthana and Hawker's, Richard's Standard nutrient and Czapek's Dox medium supported poor growth of the pathogen. Sporulation was not observed on any of the liquid media tested.

(c) Effect of different temperature on the growth and sporulation of the pathogen:

Temperature is known to have profound effect on the growth of micro-organisms. Present studies were, therefore, taken up to find out the optimum, minimum and maximum temperature requirements for the growth and sporulation of the pathogen. For this investigation, the fungus was grown at ten different temperatures ranging from 15^oC to 45^oC, and the average dry mycelial weights were recorded, and the results are summarized in Table-8 and Histogram-5.

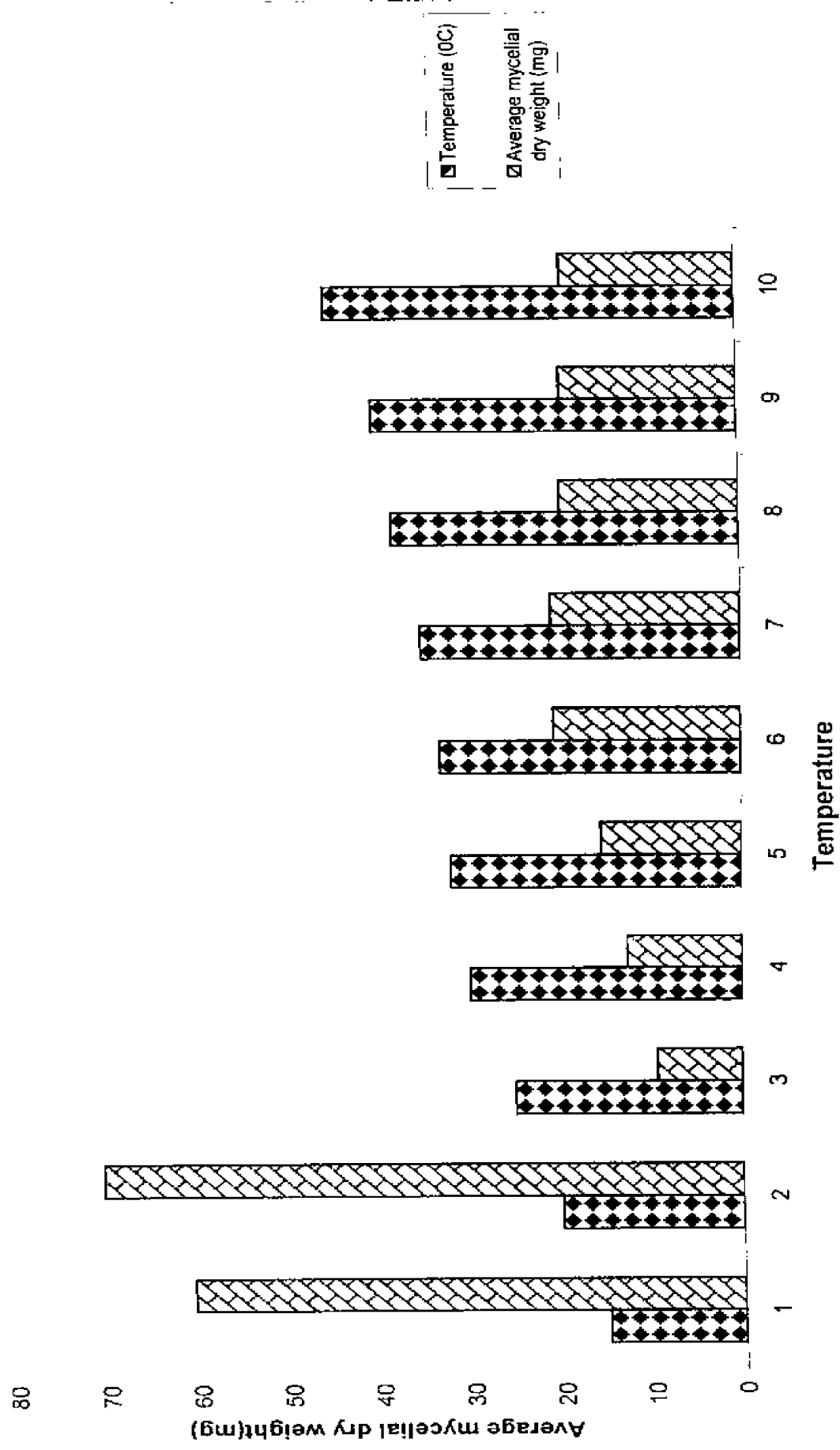
Table 8 : Effect of various temperatures on the growth and sporulation of *R. bataticola*.

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1.	15	6.62	-
2.	20	7.46	-
3.	25	9.30	-
4.	30	12.63	-
5.	32	15.43	-
6.	33	20.57	-
7.	35	20.70	-
8.	38	19.50	-
9.	40	19.27	-
10.	45	19.00	-
CD. at 5%		1.88	-

(-) Indicates no sporulation

The results (Table-8 and Histogram-5) indicate that, although, the pathogen could grow over a wide temperature range of 15^oC to 45^oC, but the optimum temperature for its growth was found to be 35^oC. The next best temperature for its

Histogram-5 : Effect of various temperatures on mycelial dry matter of *R. bataticola* in vitro .



growth 33°C, followed by 38°C and these treatments did not differ among themselves. The growth of the pathogen gradually decreased both below and above the optimum (35°C) and the minimum growth was recorded at 15°C. Sporulation of the fungus was not observed at any temperature value.

(d) Effect of different pH levels on the growth and sporulation of the pathogen:

It is well documented that hydrogen ion concentration exerts considerable influence on the growth and sporulation of microorganisms. Therefore, studies were conducted to find out the maximum, optimum and minimum pH for the growth of the pathogen, under study. In this investigation the fungus was grown on potato dextrose agar medium adjusted at pH levels ranging from 3.0 to 9.0 and the dry mycelial weight as well as sporulation were recorded and the data obtained are presented in Table-9 and Histogram-6.

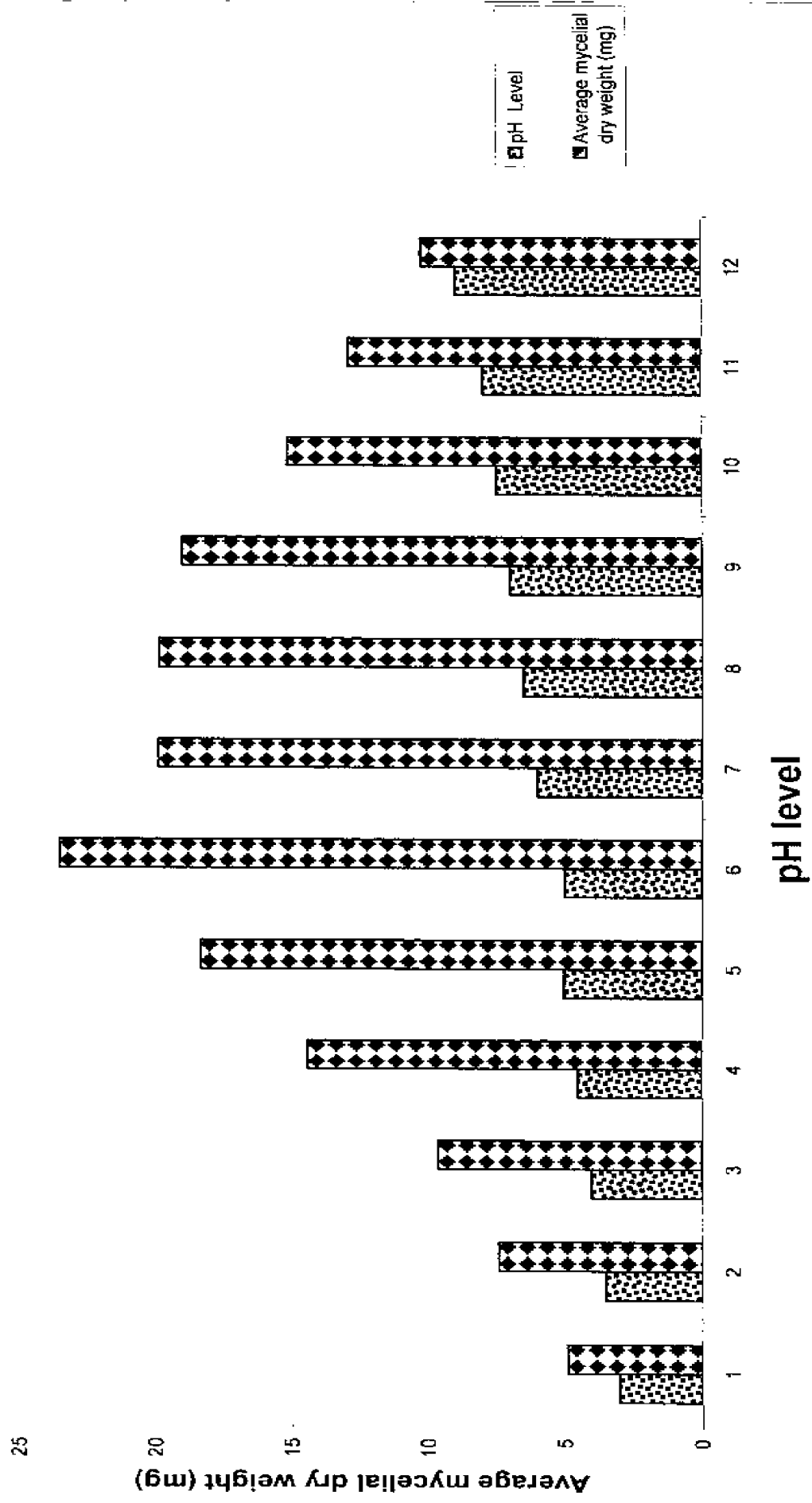
Table 9: Effect of varying pH levels on the growth and sporulation of *R. bataticola*.

S. No.	pH Level	Average mycelial dry weight (mg)	Sporulation
1.	3.0	4.89	-
2.	3.5	7.4	-
3.	4.0	9.63	-
4.	4.5	14.42	-
5.	5.0	18.36	-
6.	5.5	23.53	-
7.	6.0	19.89	-
8.	6.5	19.87	-
9.	7.0	19.00	-
10.	7.5	15.15	-
11.	8.0	12.93	-
12.	9.0	10.25	-
CD. at 5%		1.21	

(-) Indicate no sporulation observed

The results (Table-9 and Histogram-6) indicated that although the pathogen grew over a wide range of pH from 3.0 to 9.0 but the optimum pH for its growth was found to be 5.5 at which maximum growth occurred, followed by 6.0 and 6.5 and these

Histogram-6 : Effect of various pH levels on the mycelial dry matter of *R. bataticola*



two pH levels supported statistically different growth from each other. However with the increase of pH growth increased gradually with increase up to 7.0 and thereafter it declined. The minimum growth of the pathogen was recorded at pH 3.0. No sporulation of the fungus occurred at any pH value.

PRODUCTION OF TOXIN BY THE PATHOGEN *IN VITRO*

The role of toxin in pathogenesis and disease syndrome in number of fungal diseases is now well recognized. Therefore, the following investigations were carried out to detect the production of toxins by the pathogen *in vitro*.

(a) Effect of toxic metabolites on chickpea leaves

The toxic effect of metabolites produced by the pathogen *in vitro* on the chickpea leaves was investigated. Two to three drops of the culture filtrate was placed at the site of injury done with the needle on the leaf and data on development of necrotic spot around the injury were recorded thereafter which are presented in Table-10.

Table 10 : Effect of toxic metabolites of *R. bataticola* on chickpea leaves.

Treatment	Number of leaves inoculated	Number of leaves showing necrotic spot
Application of filtrate after injuring	50	50
Application of sterilized water after injuring (check)	50	0.00

It is evident from the results (Table-10) that all the leaves developed necrotic spots, these spots, after application were observed. Initially, small light brown, circular to oval spots were produced on the inoculated leaves with crude toxin of the fungus which later on increased in size and become, more or less oval to irregular and dark brown in colour. These spots did not resemble to those produced by the pathogen in host in nature.

(b) Effect of toxin on seed germination and seedling vigour :

The effect of toxic metabolites present in culture filtrate on seed germination as well as root and shoot length was studied. The results were summarized in Table-11.

Table 11 : Percentage of seed germination at various concentrations and heated and unheated culture filtrates of *R. bataticola*.

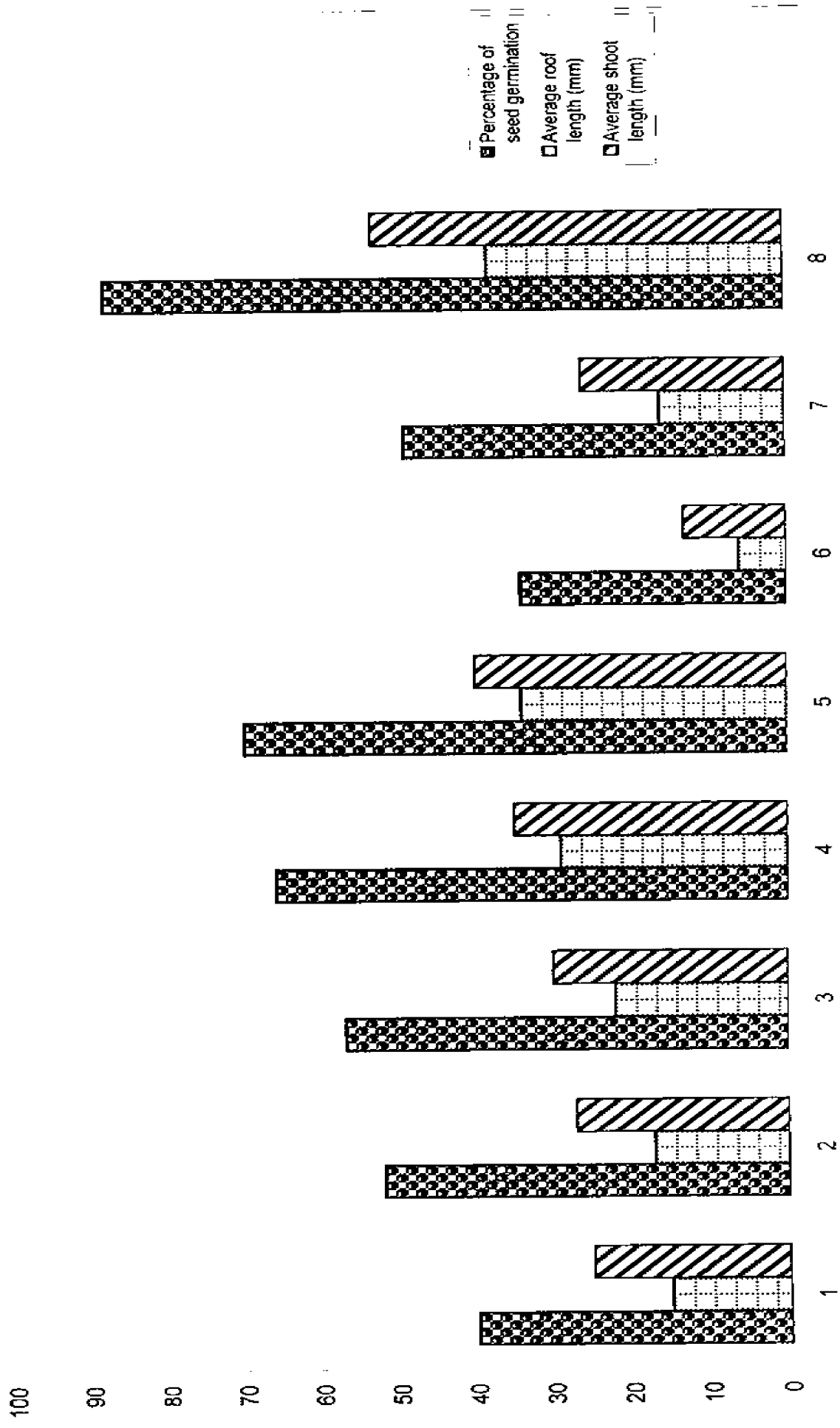
Test solution / concentration of culture filtrate	Percentage of seed germination	Average root length (mm)	Average shoot length (mm)
1:10	40	15	25
1:50	52	17	27
1:100	57	22	30
1:500	66	29	35
1:1000	70	34	40
Unheated culture filtrate (crude)	34	6	13
Heated culture filtrate (crude)	49	16	26
Sterilized distilled water	88	38	53

It is evident from the (Table-11 and Histogram-7) that there was a marked inhibition in seed germination as well as root and shoot length in case of various dilutions of the filtrate as well as heated and unheated metabolites as compared to the control. The inhibitions were more pronounced in unheated culture filtrate, followed by concentration 1:10. Such reduction, although to lesser extent, were also recorded in treatments where the seeds were soaked in heated culture filtrate. This was followed by the concentration 1:50, 1:100, 1:500 and 1:1000.

SURVIVAL OF THE PATHOGEN :

The knowledge of survival of the pathogen in nature is of paramount importance in planning the effective strategy of control in order to minimize the losses. Therefore, the role of seed, infested soil and diseased plant debris in the survival of *R. bataticola* and primary infection of the disease was studied separately.

Histogram-7 : Per cent of seed germination at various concentrations and heated and unheated culture filtrates of *R. bataticola*.



(a) Role of seed in survival of the pathogen

To determine the role of seeds in the survival of the pathogen, isolations were taken from the naturally infected seeds as well as from the healthy seeds mixed with mycelial suspension of the pathogen. The results obtained are summarized in Table-12.

Table 12: Survival of *R. bataticola* in association with chickpea seeds under the laboratory conditions

Treatments	Apr.	May	June	July	Aug.	Sep.	Oct.
Naturally infected seeds	+	+	+	+	+	-	+
Healthy seeds mixed with the mycelial suspension of <i>R. bataticola</i>	+	+	+	+	+	+	+

(+) Denoted pathogen isolated

The results presented in the Table-12 showed that naturally infected seeds as well as healthy seeds mixed with mycelial suspension of the pathogen, yielded the pathogen culture of *R. bataticola* upto seven month from April to October in respective year. Thus, it was concluded that both the naturally infected and the artificially infested seeds served as the carrier of the pathogen from one season to another

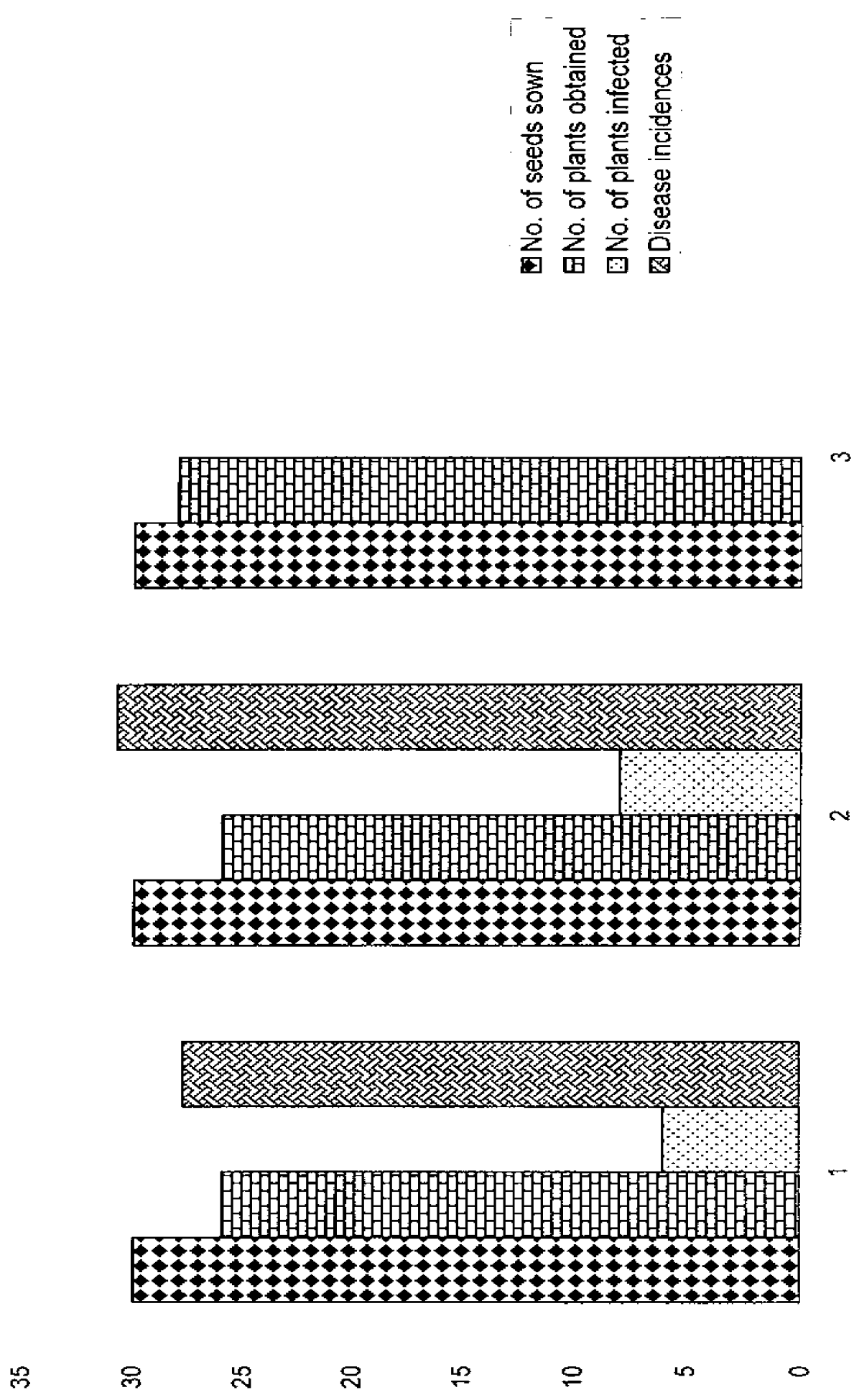
(b) Role of seed in the primary infection of the disease :

In order to ascertain the role of seed as a primary source of infection of the disease, studies were conducted and the results are presented in Table-13 and Histogram-8.

Table 13: Role of seeds in the primary infection of the disease.

Treatments	No. of seeds sown	No. of plants obtained	No. of plants infected	% Disease incidence
Naturally infected seeds sown in autoclaved soil	30	26	6	23.08
Seeds mixed with the mycelial suspension and sown in autoclaved soil	30	26	8	30.76
Sterilized healthy seeds sown in autoclaved soil (control)	30	28	-	-

Histogram-8 : Role of seeds in the primary infection of the disease.



The results (Table-13) revealed that the dry root rot developed in plant raised from the naturally infected as well as artificially infested seeds, thereby demonstrating their convincing role in primary infection of the disease. However, the percentage of plants infected was higher in case of artificially infested seeds than that of naturally infected ones. No dry root rots were observed in case of control. Thus, it was concluded that the seeds not only played significant role in the survival of the pathogen but also in the primary infection of the disease.

(c) Role of infested soil in survival of the pathogen

In order to determine the role of infested soil in the survival of the pathogen, the naturally infested field soil and autoclaved soil mixed with the fungus were used. Isolations were taken from these soils and the observations recorded are presented in Table-14.

Table 14: Survival of *R. bataticola* in the infested soil in the laboratory

Treatments	Apr.	May.	June	July	Aug.	Sep.	Oct.
Naturally infected soil	+	+	+	+	+	+	+
Autoclaved soil mixed with the fungus culture	+	+	+	+	+	+	+

(+) Denotes pathogen isolated

It is evident from the results (Table-14) that pathogen was isolated for seven months from both the soils i.e. naturally infested soil and artificially infested autoclaved soil from April to October. Thus, it remained viable in the infested soil from one season to another.

(d) Role of infested soil in primary infection of the disease:

To ascertain the role of infested soil in the primary infection of the disease, observations were recorded on plants raised in pots containing naturally infested field soil, artificially infested soil and autoclaved soil sown with sterilized seeds separately. The results are summarized in Table-15.

Table-15 : Role of infested soil in primary infection of the disease.

Treatments	No. of seeds sown	No. of plants obtained	No. of plants infected	% Disease incidence
Sterilized seeds sown in naturally infested field soil	30	26	6	23.08
Sterilized seeds sown in autoclaved infested soil	30	27	9	33.33
Sterilized seeds sown in autoclaved soil	30	28	-	-

It is apparent from the results in (Table-15 and Histogram-9) that the plants raised in the pots filled with naturally infested field soils and artificially infested soil produced dry root rot. However, the percentage of plant infection was more in case of artificially infested soil than naturally infested field soil. No dry root rot was observed in case of control where sterilized seeds and soil were used. This clearly showed that the pathogen was not only able to survive in infested soil from one season to another but also played an important role as the source of primary infection of the disease.

(e) Role of diseased plant debris in survival of the pathogen :

(a) Under laboratory condition : For this study, the affected roots by *R. bataticola* of the highly susceptible chickpea variety JG-62 were collected and stored in the laboratory and isolations were taken from this material to ascertain the survival of the pathogen till October (Table-16).

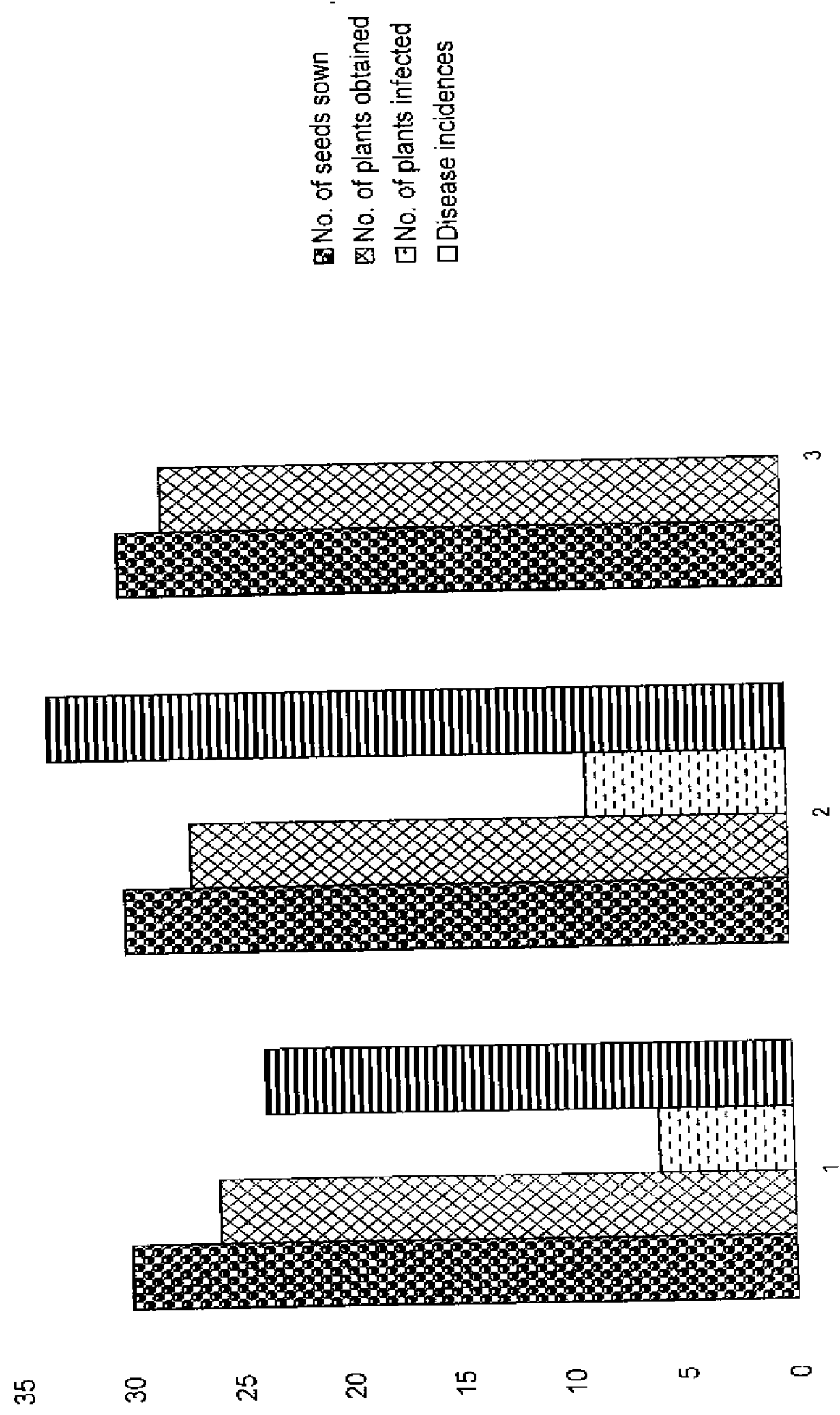
Table 16 : Survival of *R. bataticola* in diseased plant debris.

Treatments	Apr.	May.	June	July	Aug.	Sep.	Oct.
Diseased plant debris stored in the laboratory (15°C – 38°C)	+	+	+	+	+	+	–
Diseased plant debris kept buried in pot soil in the open (10°C – 49°C)	+	+	+	+	+	+	–

(+) Denotes pathogen isolated

It was observed that the pathogen was able to survive for a period of seven months till the next crop season in all cases, pure culture of *R. bataticola* was also obtained.

Histogram-9 : Role of infested soil in primary infection of the disease in field.



(b) **Under field condition:** Diseased roots wrapped in tissue paper were buried in the pot soil at a depth of 8.0 cm and kept in the open. Subsequently, their role in survival of the pathogen was ascertained by isolations from these affected root parts. The results obtained are summarized in Table-16. The results revealed that the pathogen was isolated from diseased root parts of the plants in infection stage from April to October.

(f) Role of plant debris in primary infection of the disease :

In order to ascertain the role of diseased plant debris as a primary source of infection the experiment was conducted as described under "Materials and Methods". The observations on the development of dry root rot were recorded and the results are presented in Table-17.

Table 17 : Role of diseased plant debris in primary infection of the disease.

Treatments	No. of seeds sown	No. of plants obtained	No. of plants infected	% Disease incidence
Sterilized seeds sown in autoclaved soil mixed with disease plant debris kept in the laboratory	30	25	4	16
Sterilized seeds sown in autoclaved soil mixed with disease plant debris kept in open (buried in pot soil)	30	26	3	11.54
Sterilized seeds sown in autoclaved soil (control)	30	28	-	-

It is clear from the results (Table-17 and Histogram-10) that the chickpea plants raised in pot soil infested with the diseased plant debris kept at room temperature, developed dry root rot. This clearly indicated that the diseased plant debris stored at room temperature played significant role in survival of the pathogen and primary infection of the disease. Similarly, the disease development also occurred in case of plant raised in pot soil mixed with the diseased plant debris, kept in open. However, the dry root rot was more in case of diseased plant debris kept in the laboratory than that

kept in open. Thus, it was concluded that diseased plant debris kept under field conditions also played a significant role in the primary infection of the disease.

HOST RANGE OF THE PATHOGEN :

Information on the host range of the pathogen is of immense value for understanding the mode of its survival during the off-season. The role of other hosts in spread of the disease during the crop season as well as the recurrence of the disease is equally important. Ultimately this aids in the planning of successful disease management. Therefore, in the present study 71 plants species including crop plants and weeds belonging to 17 different families, were subjected to artificial inoculation to find out the range of infectivity of the pathogen. The results obtained and presented in Table-18.

Table 18 : Host range of *R. bataticola* under artificial conditions of inoculation.

S. No.	Family	Plant species	Disease reaction
1	Araceae	<i>Colocasia esculenta</i> L.	+
2	Caricaceae	<i>Carica papaya</i> L.	+
3.	Chenopodiaceae	<i>Beta vulgaris</i> L.	+
		<i>Chenopodium album</i> L.	-
		<i>Spinacia oleraceae</i> L.	-
4.	Compositae	<i>Ageratum conyzoides</i> L.	-
		<i>Carthamus tinctorius</i> L.	-
		<i>Chrysanthemum indicum</i> L.	+
		<i>Dahlia variabilis</i> L.	+
		<i>Helianthus annuus</i>	+
		<i>Lactuca sativa</i>	-
		<i>Tagetes erecta</i>	-
5	Convolvulaceae	<i>Ipomoea batatas</i> L.	+
6.	Cruciferae	<i>Brassica campestris</i> L.	-
		<i>Brassica juncea</i> L.	-
		<i>B. oleracea</i> var. <i>botrytis</i> L.	-
		<i>B. oleracea</i> var. <i>capitata</i> L.	-
		<i>B. rapa</i> L.	-
		<i>Raphanus sativus</i> L.	+

7.	Cucurbitaceae	<i>Cucumis melo</i> L. <i>C. sativus</i> L. <i>Cucurbita maxima</i> . <i>Lagenaria vulgaris</i> Ser. <i>Luffa cylindrica</i> Roens. <i>Momordica charantia</i> L. <i>Citrulus vulgaris</i> L. <i>Trichosanthes dioica</i> Roxb.	+ + + + + + + +
8.	Euphorbiaceae	<i>Euphorbia hirta</i> L. <i>Ricinus communis</i> L.	- +
9.	Gramineae	<i>Avena sativa</i> L. <i>Cynodon dactylon</i> L. Pers. <i>Eleusine coracana</i> L. <i>Hordeum vulgare</i> L. <i>Oryza sativa</i> L. <i>Pennisetum typhoides</i> <i>Setaria italica</i> <i>Sorghum vulgare</i> <i>Triticum aestivum</i> L. <i>Zea mays</i> L.	- - + - + - + + - +
10.	Liliaceae	<i>Allium cepa</i> L. <i>Allium sativum</i> L.	- -
11.	Linaceae	<i>Linum grandiflorum</i> Desf. <i>Linum usitatissimum</i> L.	- -
12.	Malvaceae	<i>Abelmoschus esculenta</i> L. Moench <i>Althoea rosea</i> L. <i>Gossypium hirsutum</i> L. <i>Cyamopsis tetragonoloba</i> L.	- - + -
13.	Papilionatae	<i>Arachis hypogea</i> L. <i>Cajanus cajan</i> (L.) Millsp. <i>Dolichos lablab</i> L. <i>Glycine max</i> (L.) Merr. <i>Lens esculenta</i> <i>Pisum sativum</i> L.	+ + + + + +

		<i>Trigonella foenumgracecum</i> L.	+
		<i>Vigna radiata</i> (L.) Wilczek	+
		<i>Vigna sinensis</i> (L.) Endl.	+
		<i>Phaseolus aureus</i>	+
		<i>Alhagi camelorum</i>	+
		<i>Desmodium trifolium</i>	+
		<i>Vicia hirsuta</i>	+
		<i>Vicia sativa</i>	+
14.	Pedaliaceae	<i>Sesamum indicum</i> L.	+
15.	Solanaceae	<i>Capsicum annuum</i> L.	+
		<i>Datura fastuosa</i> L.	+
		<i>Lycopersicum esculentum</i> Mill	+
		<i>Solanum melongena</i> L.	+
		<i>Solanum nigrum</i> L.	+
		<i>Solanum tuberosum</i> L.	+
16.	Tiliaceae	<i>Corchorus capsularis</i>	+
17.	Umbelliferae	<i>Coriandrum sativum</i> L.	+
		<i>Daucus carota</i> L.	+

(+) Indicates infected

(-) Indicates not infected

The results (Table-18) revealed that out of 71 plant species tested the pathogen was able to infect 46 of them. It failed to infect only 25 plant species, which clearly indicated that the fungus has a very wide host range.

It was observed that *R. bataticola* was able to parasitized on majority of the hosts of families viz. Papilionatae, Cucurbitaceae, Solanaceae, Compositeae and Gramineae. All the plants of family Papilionatae to which chickpea also belongs, got infected upon artificial inoculation. None of the plant species belonging to family Liliaceae and Linaceae got infected under artificial inoculation. The pathogen was isolated from those inoculated hosts in which the dry root rot were produced.

SCREENING OF CHICKPEA GERMLASM FOR DISEASE RESISTANCE:

A total of 45 varieties/cultures of chickpea were screened for varietal reactions against the disease by adopting Blotter paper technique (Nene *et al.*, 1981). The

varieties/ cultures, found resistant and moderately resistant were again screened in pots under conditions of artificial epiphytotic of the disease in pots, replicated three times. The results obtained in this study, are presented in Table-19.

Table 19: Reaction of different varieties/ cultures of chickpea to *R. bataticola*

Categories of resistance / susceptibility	Varieties/Cultures
Resistant	H 355, JG 315, Pusa 212, G543, ICC 32, BG 244, Phule G 5, ICC 2644, 10384, 11224, 10630, 11332, ICCV10, ICC 12441, 12263, ICCL 81010, 81002, Avrodhi, ILC 702, 200,
Moderately resistant	GCP- 9504, Phule G 96313, 96105, 96020, GL 91059
Tolerant	JG-9, 22,23, Phule G 96022, 95123, 93208, 92028, BCP 91, 28
Moderately susceptible	GC 90178, GCP 9605, 9518, IPC- 95-1, IPCK-96-1, Saki 93130. Radhey.
Susceptible	Vijay, GCP 9610, GL91061, JG-62

It is evident from the results (Table-19) that out 45 varieties/ cultures, tested by Blotter paper technique and pot experiment, twenty viz., H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G5, ICC 2644, 10384, 11224, 10630, 11332, ICCV 10, ICC 12441, 12263, ICCL 81010, 81002, Avrodhi, ILC 702 and 200, proved resistant; five namely, GCP- 9504, Phule G 96313, 96105, 96020 and GL 91059 moderately resistant, nine viz., JG-9, 22,23, Phule G 96022, 95123, 93208, 92028, BCP 91 and 28, tolerant and the rest were found to be moderately susceptible and susceptible. (Fig.-13 & Fig -14).

MANAGEMENT OF THE DISEASE :

Screening of the fungitoxicants against the pathogen in bioassay test

In order to select the most effective fungitoxicants, 10 fungicides, 8 non-target pesticides (5 insecticides, 3 herbicides and 1 antibiotic), and 2 bio-agents all together, 21 fungitoxicants belonging to different groups, were screened in the laboratory using the technique Schmitz (1930).



Fig. 13

Plate -VIII

Fig.14: A susceptible line inoculated and moderately resistant line showing the extent of infection using Blotter paper technique.

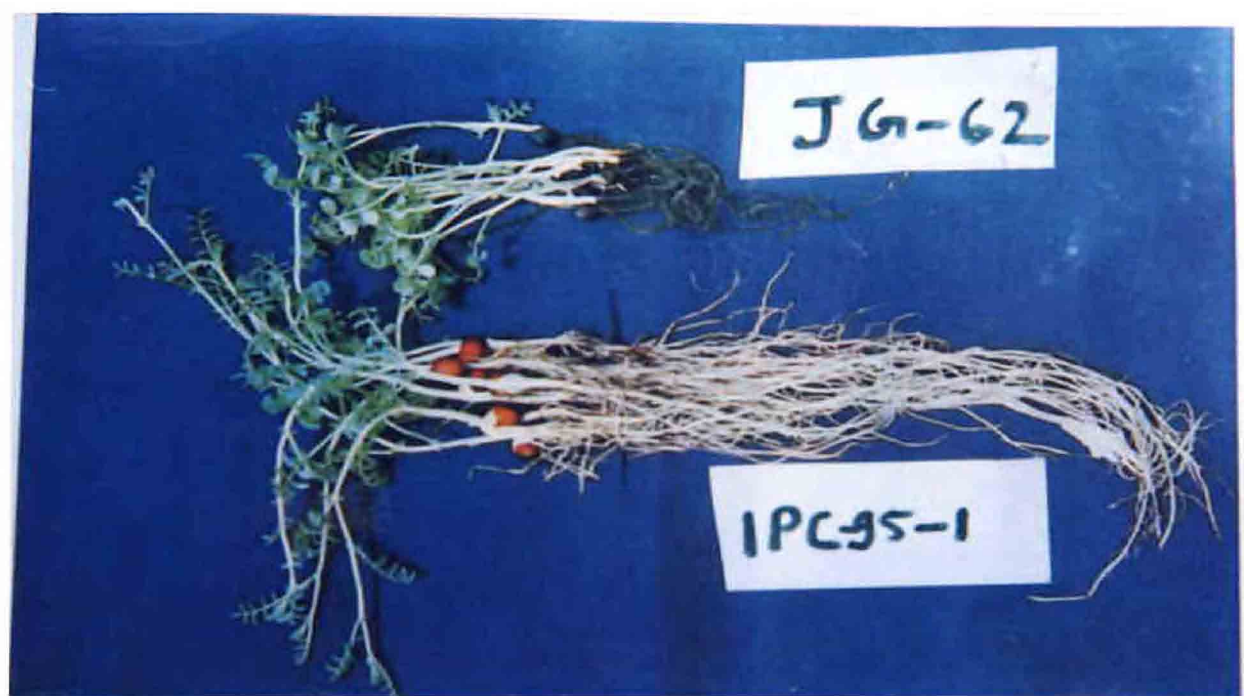


Fig. 14

Table 20 : Inhibitory effect of different group of fungicides, non-target pesticides and bio-agents on the growth of *R. bataticola* in bioassay test on P.D.A.

Fungitoxicants	Doses (ppm)	Average diameter of fungal colony (mm)	Per cent inhibition over control
A. Fungicides			
Carbendazim 25 DS	2500	00.00	100.00
Carbendazim 50 WP + Thiram 80 WP	2500	00.00	100.00
Carboxin 75 WP	2500	00.00	100.00
Topsin-M 50 WP	2500	00.00	100.00
Thiram (1:1) 80WP	2500	07.00	92.09
Mancozeb 80WP	2500	09.00	89.83
Captan 50 WP	2500	10.00	88.70
Thiobendazole 60 WP	2500	10.00	88.70
Ziram 80 WP	2500	32.00	63.84
Blue copper-50	3000	69.00	22.03
B. Non-target pesticides			
i. Antibiotic			
Streptocycline	2500	61.00	31.07
ii Insecticides			
Monocrotophos 40 EC	500	11.00	87.57
Indosulfan 35FC	700	14.00	84.18
Phorate 10G	25000	16.00	81.92
Chloropyrifos 20 EC	300	20.00	77.40
Phosphomidan 85 EC	500	25.00	71.75
iii Herbicides			
Lasso	1200	05.00	94.35
Basalin	1200	55.00	37.85
Isoproturan	1200	60.00	32.20
C. Bio-agents			
<i>Bacillus subtilis</i>	*	20.00	77.40
<i>Funginil (Trichoderma harzianum)</i>	**	27.00	69.49
Control	-	88.50	-

CD at 5% level 1.458

* 10^6 cfu/gm

** 20×10^6 cfu/gm

The growth of the fungus in various treatments was measured and the average diameter of the colony in each group was recorded along with that of control and the results are presented in (Table-19 and Histogram-11).

It is evident from the results Table-19 that out of 21 tested fungitoxicants, four Carbendazim 25 DS, Carbendazim 50 WP + Thiram 80 WP, Carboxin 75 WP and Topsin –M 50 WP proved to be the most effective as they inhibited the fungal growth completely. The remaining 17 fungitoxicants, although inhibited the growth of fungus to varying degrees, but failed to inhibit completely and therefore, they were considered to be partially effective. Among the partially effective. Lasso gave the highest inhibition of the fungal growth and was significantly superior to the remaining once. Thiram (1:1) 80 WP observed to be the next best, followed by Mancozeb 80 WP, Captan 50 WP and Thiobendazole 60 WP, in which Captan 50 WP and Thiobendazole 60 WP did not differ significantly from each other and Monocrotophos 40EC was at par with Indosulfan 35EC in checking the fungal growth. The rest of the fungitoxicants in the descending order of efficacy were Phorate 10G, Chloropyrifos 20EC *Bacillus subtilis*, Phosphomidan 85EC, Funginil (*Trichoderma harzianum*) Ziram 80WP, Basalin, Isoproturan, Streptocycline and Blue copper-50. Streptocycline and Blue copper-50 were found to be statistically near about similar in their efficacy.

Efficacy of seed dressing fungicides on seed germination and seedling infection in pots experiment :

Eight-seed dressing fungicides, which proved effective in bioassay test, were selected to assess their efficacy in improving the seed germination and controlling seedling infection in pot experiment. Chickpea seeds of variety JG-62, obtained from diseased crop, were treated separately with various seed dressing fungicides. These seeds were sown in quadreplicate pots per treatment along with untreated control. The data on seed germination and seedling infection were recorded and results of the experiment are presented in Table-21.

Histogram-11: Showing inhibitory effect of different fungitoxicants

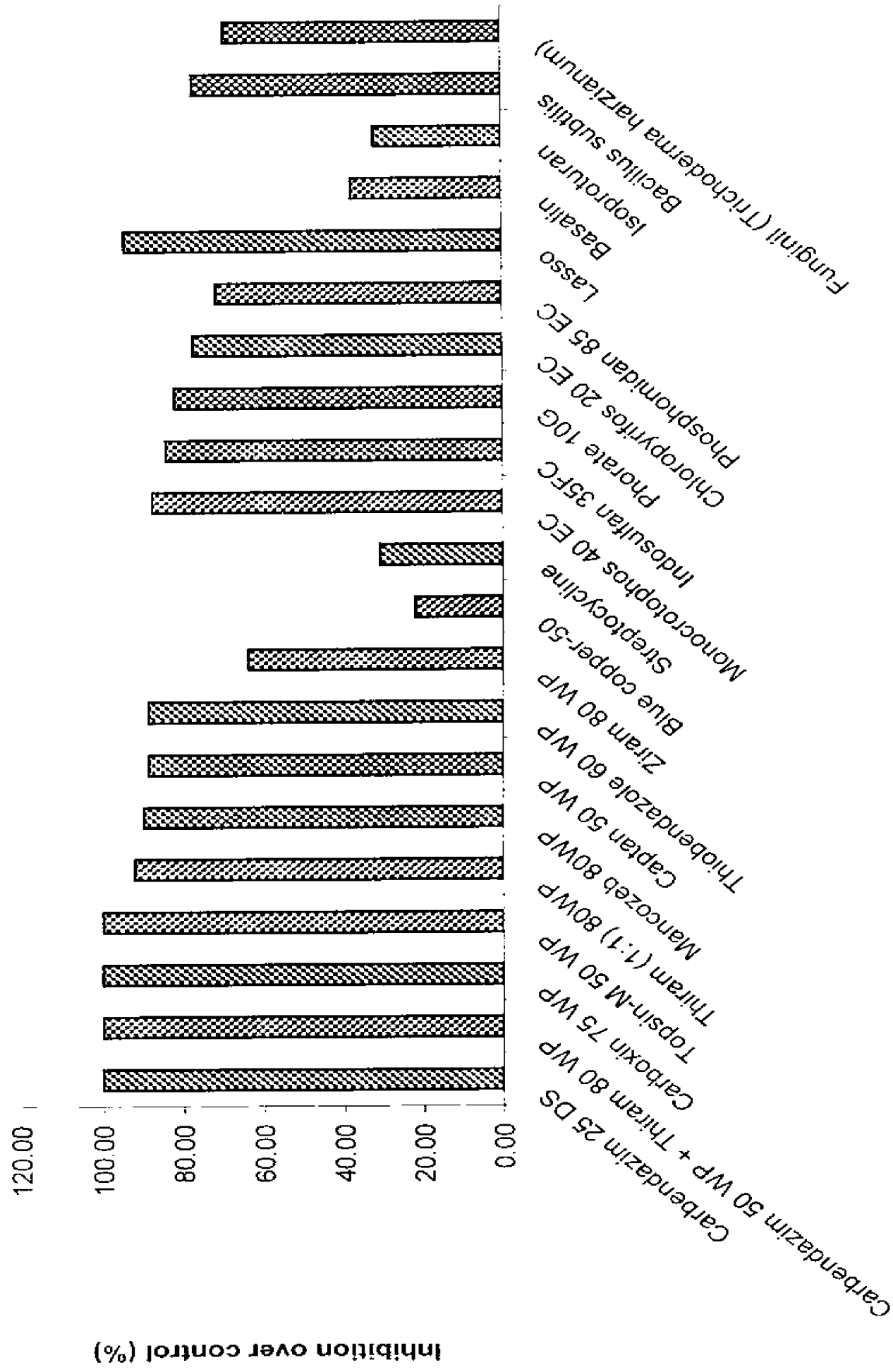
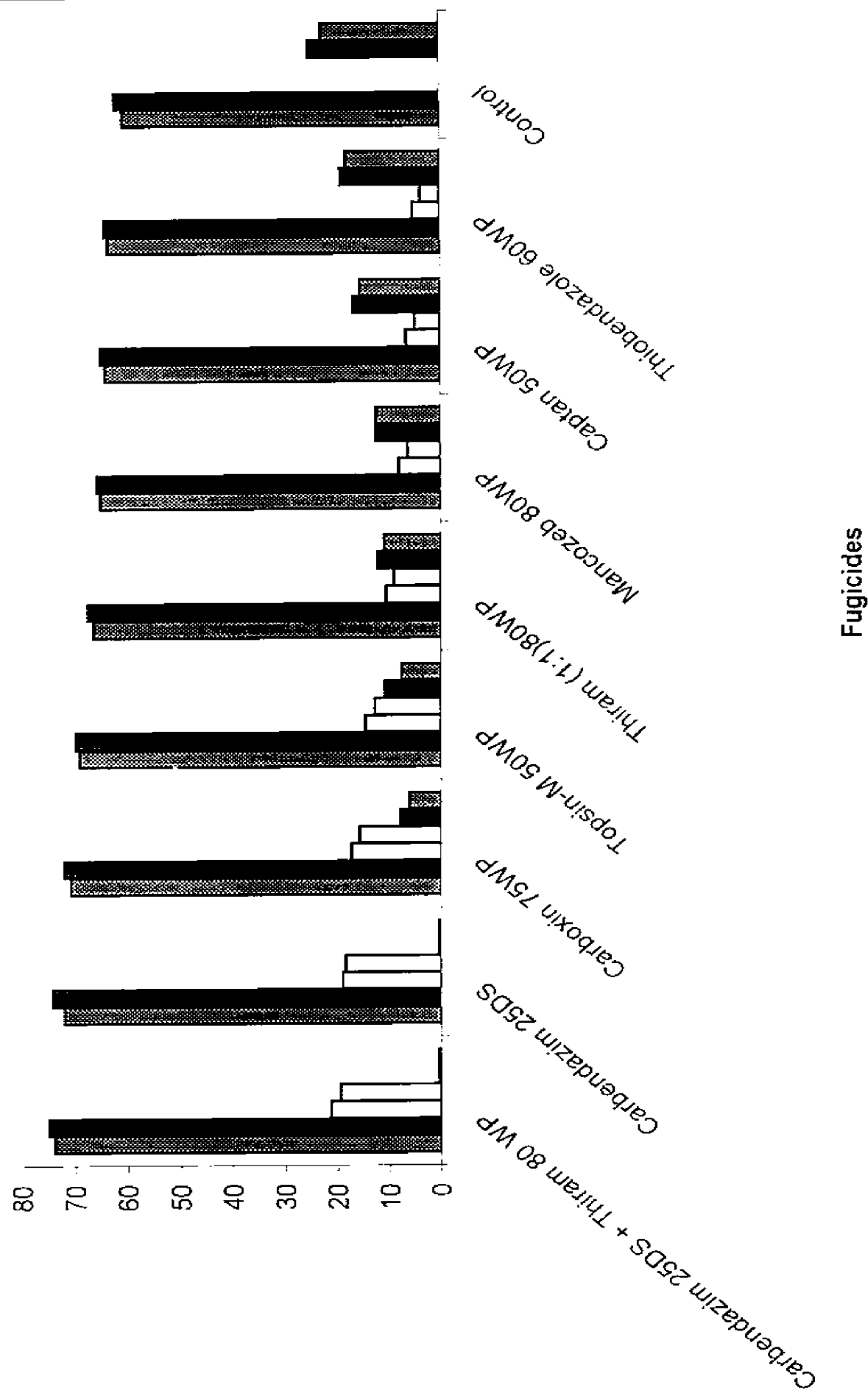


Table 21: Effect of seed treatment with fungicides on seed germination and seedling infection of chickpea caused by *R. bataticola* in pot experiment.

S. No.	Treatment	Dose (%)	Seed germination corresponding value (%)		increase in germination (%)over control		Seedling infection value (% corresponding value)	
			1998-99	1999-2000	1998-99	1999-2000	1998-99	1999-2000
1.	Carbendazim 25DS + Thiram 80 WP	0.2	73.83 (92.24)	75.05 (93.34)	21.28 19.42		0.59 (0.01)	0.59 (0.01)
2.	Carbendazim 25DS	0.2	72.04 (90.49)	74.22 (92.60)	18.98 18.47		0.59 (0.01)	0.59 (0.01)
3.	Carboxin 75WP	0.1	70.82 (89.20)	72.04 (90.49)	17.29 15.77		7.86 (1.87)	6.15 (1.28)
4.	Topsin-M 50WP	0.2	69.03 (87.10)	69.86 (88.14)	14.64 12.76		10.86 (3.55)	7.70 (1.79)
5.	Thiram (1:1)80WP	0.2	66.50 (84.09)	67.46 (85.30)	10.57 9.13		12.19 (4.45)	10.91 (3.64)
6.	Mancozeb 80WP	0.5	65.06 (82.21)	65.75 (83.13)	8.09 6.35		12.48 (4.66)	12.48 (7.12)
7.	Captan 50WP	0.20	64.23 (81.09)	64.92 (82.09)	6.62 4.95		16.72 (8.27)	15.48 (7.12)
8.	Thiobendazole 60WP	0.20	63.48 (80.06)	64.23 (81.09)	5.27 3.74		19.16 (10.77)	18.06 (9.61)
9.	Control		60.70 (76.05)	62.14 (78.16)	-- -		25.20 (18.12)	22.83 (15.05)
	CD at 5% last		4.90	5.64			8.89	8.17

Histogram-12: Effect of seed treatment with fungicides on seed germination and seedling infection of chickpea caused by *R.bataticola* in pot experiment



The results (Table-21 and Histogram-12) revealed that among the eight seed dressing fungicides tested, Carbendazim 25 DS and Carbendazim 25 DS + Thiram 80 WP were capable of almost completely checking the seedling infection of *R. bataticola* as well as improved the seed germination. Thereafter fungicides viz., Carboxin 75 WP, Thiram (1:1) 80 WP, Mancozeb 80 WP and Captan 50 WP were also found effective in reducing the seedling infection and improving the seed germination significantly. However, the remaining fungicides, namely Topsin-M 50 WP and Thiobendazole 60wp failed to control seedling infection of chickpea as they were at par with the untreated check (control).

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DISCUSSION

DISCUSSION

Chapter-5

Chickpea (*Cicer arietinum* L.) is an important pulse crop grown in India, having the high nutritive value and manifold uses. It is necessary to improve the average yield of this crop because in comparison to many other countries of the world, its production is low. In addition to other causes, the main reason for its low production is the incidence of the diseases which cause huge losses almost every year. Therefore, special emphasis has to be given to tackle the already known and new diseases in addition to the adaptation of other package of practices. Since dry root rot of chickpea, which has posed a serious problems during the recent years in Uttar Pradesh about which little information is available so far, to fill up the lacuma in our knowledge for making effective strategies to overcome this malady, the investigation have been carried out in detail

Investigations on the prevalence and incidence of the disease showed that the dry root rot of chickpea was widly prevalent in different agro-climatic regions of U.P. during 1998-99 and 1999-2000. The disease incidence varied from 19 to 42 per cent of the different locations of U.P. Incidence (42%) of this disease under natural conditions were recorded at the place of Jaspura (Banda) followed by Mauranipur (Jhansi) and Talbehat (Lalitpur). The lowest incidence (19%) of the disease were observed at Udham Singh Nagar and Meeraganj (Bareilly). The maximum incidence of was observed in bold seeded varieties followed by medium seeded while the lowest the disease disease incidence was recorded on small seeded or *deshi* variety of chickpea.

In India, the first report of the fungus was made by **Butler and Bisby (1931)** from Madras in cluster bean, lady's finger and bringal. The first report of the dry root rot on chickpea was made by **Dastur (1947)** from West Bengal. The incidence of the dry root rot were also reported, **Tripathi and Sharma (1983)** from Haryana and **Gangwar (2000)** from differnet agro-climatic region of U.P. and found 19-36 per cent incidence surveyed of 10 locations. Except these reports there is not any other information regarding the occurrence of this disease in the field from other parts of the country.

Studies on the disease symptomatology revealed that under natural condition it appeared on the roots. The dry root rot symptoms were sudden and premature drying of the infected plants and when such plants were uprooted their root were rotten shredded. The finer roots were mainly affected and have dark and blackened streaks under beneath their bark with evident dark and sclerotial bodies. Such roots were brittle and break when touched. Rotting of root extends to the base of the stem of the plant. The lesions coalesce and cause whole plant to dry up and ultimately die. Dark brown to black pycnidia were developed on infected roots. The symptoms were which resembled with symptoms as described by **Chattopadhyay and Bhattacharya (1967)**, **Rai and Singh (1973)**, **Ilyas and Sinclair (1974)** and **Rangaswami (1994)**.

The pathogenicity of the all 12 isolates of the pathogen was conducted on healthy potted plants of chickpea and it produced characteristic symptoms of the dry root rot those observed in nature. The pathogen was re-isolated from the infected roots and was found identical to the original isolates. Thus, its pathogenicity was proved in accordance with the **Koch's postulates (1883)**. The pathogenicity of this fungus was established on chickpea and other crops has been proved by earlier workers (**Dastur, 1947** **Kulkarni and Patil, 1968**; **Dhingra and Sinclair, 1973** and **Dhingra and Sinclair, 1978**) also and their findings support the pathogenic nature of this fungus as observed in the present case.

The morphological and cultural characters of the fungus were studied on potato dextrose agar medium for identification and it was found that the light to dark gray coloured colonies with fluffly mycelium developed on potato dextrose agar medium. Mycelium was initially grayish white, became brown to dark brown with age, septated, profusely branched 4.5-7.5 μm in width in the xylem of the root, which was 3.5-7.0 μm on agar culture medium. Sclerotia were somewhat hairy owing to immergence of dark hyphae from their surface, measured 70.5-9.50 μm in diameter. Pycnidia were measured 80.5 to 110.5 μm in diameter. Pycnidiospores (conidia) were single celled hyaline, oval, elongated to elliptical in shape, oozing out in the form of dull white cirrus and measured 15.5-21 $\times$ 5.5-9.0 μm . The morphological and cultural characters of the pathogen observed in the present study are more or less, similar to those described by *R. bataticola* (Taub.) Butler/*M. phaseolina* (Tassi.) Goid as reported by

previous workers **Ashby (1927)**, **Haigh (1930)**, and **Byadgi and Hegde (1985)**, and it was identified as such *R. bataticola* (Taub.) Butler.

The effect of eight different solid media was studied on the growth of the pathogen and it was found that potato dextrose agar medium supported the best growth. Good growth was recorded on standard nutrient agar medium and chickpea root extract agar medium; for fair on carrot root extract agar, whereas the remaining media viz., oat meal agar, Czapek Dox agar, Richard's agar and Asthana and Hawker's agar supported poor to very poor growth. More or less similar results were obtained with these media in liquid form also. No sporulation was observed on any of the solid or liquid media tested. Earlier, no such systematic effort was made to find out the best suitable medium for the growth and sporulation of *R. bataticola*. However, with few exceptions, the reports on media studies made for *R. bataticola* by some workers (**Post, 1933**; **Livingston 1945**; **Goth and Ostazisk, 1965**; **Knox-Davies, 1965**) more or less supported the present findings. In the present study no sporulation of the fungus was recorded on any of the media tested, when subcultured culture was used. The non-sporulating nature of this fungus was emphasized by **Knox-Davies (1965)** also who reported that sporulation could not included unless the cultures were irradiated with ultra-violet light for 12 to 24 hours at 30°C. This report neglected by the observations of **Goth and Ostazisk (1965)** who induced sporulation in 14 isolates at 30°C. All workers suggested that glucose containing medium could supported the growth. The growth of this fungus on potato dextrose agar medium standard nutrient agar medium and chickpea root extract agar media including liquid forms appears to constitute the new report.

In studies taken up to find out the an optimum temperature for growth and sporulation of the pathogen it was observed that *R. bataticola* could grow over a wide range of 15°C to 45°C and the optimum temperature for its growth was found to be 33°C followed by 32°C and 35°C. Sporulation of the fungus was not observed at any temperature. These results agree closely with the findings of **Jackson (1965)** who reported that maximum growth and day night occurred at 26 to 32°C but the isolates varied in the total amount of growth at optimum temperature requirement of *R. bataticola* were obtained by other workers (**Vasudeva, 1936**; **Uppal et al.; 1936**; **Adam and Stokes, 1942** and **Smith, 1964**) Non-sporulation of the fungus in the

present study may be explained due to the fact that specific treatments for sporulation were not applied.

Investigation on the pH requirement of the pathogen revealed that although, it could grow over a wide pH range of 3.0 to 9.0, but the optimum pH for its growth was found to be 5.5. The growth of the fungus was reduced both below and above the optimum pH value. However, no sporulation of the fungus occurred at any pH level tested. These results are collaborated by the findings of **Dhingra and Sinclair (1973)**, who reported that pH 5.5 favoured good growth of *M. phaseoli* except for slight variation, the observations of **Bhargava (1965)** regarding pH requirement of this fungus are also in line with the present finding. Non –sporulation of the fungus in the present study was recorded probably due to the fact that precise treatments for sporulation were not applied.

In studies, carried out to find out the production of toxic metabolites by the pathogen *in vitro*. It was observed that this fungus produced certain toxic metabolites in medium which, when placed on host leaves, produced necrotic spots, these spots were not to similar to the natural symptoms of the disease. These findings are in conformity with the observations of **Mathur (1968)** who reported that the culture filtrates of *M. phaseolina* produced typical foliar symptoms on the leaves of sunflower. **Chan and Sackston (1969,1973)** also reported that culture filtrate of *M. phaseolina* developed necrotic spots on the leaves of sunflower. They also reported that toxin was not host specific and spots related to the virulence of the isolates. Again in 1973, **Rai and Singh**, reported that toxic substance produced in the culture of *M. phaseolina* caused necrosis, chlorosis and wilting of *Brassica juncea* leaves and inhibited seed germination. **Dhipanwita - Bhattacharya et al (1992)** isolated Asperlin, Isoasperlin and Acetylphomalacetone from culture filtrates of *M. phaseolina* and reported that these three metabolites and the previously isolated 'Phaseolinone' and Phaseolinic acid caused non-specific leaf necrosis on several plants. The phytotoxic effect of culture filtrate of *R. bataticola* on chickpea leaves is being reported here for the first time.

The study on the effect of culture filtrate of the fungus on seed germination of chickpea indicated that it inhibited seed germination and shoot length of the seedlings at various concentrations. The heated culture filtrate also caused inhibition of seed germination as well as root and shoot length, although, the inhibition was more

pronounced in case of unheated culture filtrate. This indicated that the inhibitory substance in the filtrate was only partially thermostable. Hetherto no more information is available on the inhibition of seed germination as well as shoot and root length of chickpea by the metabolites of *R. bataticola* in culture filtrates, except for one report by **Dipanwita-Bhattacharya (1994)** who obtained phaseolinone, a phytotoxic metabolite of *M. phaseolina* in infected *Vigna mungo* and correlated well with germination inhibition and amount of toxin produced; 50% inhibition was observed at a toxin level of 2.1 eg/g of wet tissue. However, the report of inhibition of seed germination of *Parthenium argeatum* and *Vigna mungo* by culture filtrates of *M. phaseolina* was observed **Dhipanwita-Bhattacharya et al., (1992)** and **Kuti et al., (1997)**

In studies, regarding the role of seeds in the survival of the pathogen and disease perpetuation, it was observed that the fungus could be isolated from both naturally infected as well as artificially infected seeds from April to October, till the next crop season. The seeds, when sown in the following season, developed the dry root rot symptoms. The role of chickpea seeds in the survival of *R. bataticola* and inhibition of the disease has not been studied before. However, the present findings are supported by work of **Andrus, 1938; Lal and Mathur; 1967; Sackston, 1969** and **Nicholson and Sinclair, 1973.**

Investigation on the role of infested soil in the survival of the pathogen, revealed that it could be isolated from infested soil for sever months, till the next sowing season and the plants raised in pots, filled with ingested soil, produced typical symptoms of the disease indicating thereby that infested soil served as a source of primary inoculum for the pathogen. Some work has been done before to study the role of infested soil in survival of this pathogen of chickpea. **Taya et al., (1988)** reported that incidence of *R. bataticola* on chickpea was higher on sandy than on clay soil how soil moisture was conducive to disease development by the reports of **Collins et al., (1991), Rangaswami (1994) and Singh (1998)** concluded that the infested soil played an important role as the source of primary inoculum of the disease, which is supports the present findings.

In studies, regarding the role of diseased plant debris in the survival of the pathogen and perpetuation of the disease revealed that the pathogen could survive in the diseased plant debris stored in the laboratory as well as kept buried in pot soil in

open, for seven months till the next sowing season. The plants raised in pot soil mixed with the diseased plant debris of both the conditions separately developed typical disease symptoms. Thus, the diseased plant debris was found to play a positive and significant role not only in the survival of the pathogen but also in the primary infection of the disease. No work appears to have been done before to find out the role of diseased plant debris in the survival of the pathogen and inhibition of the disease on chickpea. However, **Norton (1953)** and **Byadgi and Hegade (1985)** also reported about survival of pathogen in plant debris. All these findings provide credence of present findings.

The host studies of the pathogen were carried out, using 71 plant species belonging to 17 families, under artificial inoculation. The pathogen was capable to cause infection in 46 hosts and only 25 plants species showed negative reaction. It was further observed that all the plants of family Papilionataeto which chickpea also belongs, got infected upon artificial inoculation *R. bataticola* was able to parasitized majority of the hosts of families viz.; Papilionatae Cucurbitaceae, Solanaceae, Compositae and Gramineae. None of the plants species belong to family Liliaceae and Linaceae got infected under artificial inoculation. These findings are in conformity with the results of **Philip et al., (1969)** who reported that *R. bataticola* infected *Phaseolus aureus* **Singh et al., (1990)** reported 10 new host of pathogen isolated from chickpea. **Songa and Hillocks (1996)** also reported that all the legume crops and weeds tested become infected after inoculation with *M. phaseolina*. *P. vulgaris*, soybean and cowpea were the most susceptible and pigeonpeas, green gram and lablab were moderately susceptible. Groundnut, chickpea were least susceptible. *Alhagi camelorum*, *Desmodium trifolium*, *Vicia hirsuta* and *Vicia sativa*, all weeds belonging to Papilionatae have not reported earlier. Therefore, these weeds host of *rabi* season may be new host report of the pathogen. The results of all these workers corroborate the results of present finding.

Investigation were carried out to located the sources of resistance against *R. bataticola* by subjecting 45 varieties/cultures of chickpea to testing first by blotter paper technique(Nene *et al.*, 1981) followed by, followed by pot experiment testing under epiphytotic conditions. Out of these genotypes twenty viz., H355, JG315, Pusa 212, G543, ICC32, BG244, Phule G5, IICC 2644, 10384, 11224, 10630, 11332,

ICCV 10, ICC12441, 12263, ICCL 81010, 81002 Avrodhi, ILC 702 and 200 proved resistant; five namely; GCP-9504, Phule G 96313, 96105, 96020 and GL 91059 moderately resistant, nine viz., JG-9, 22, 23, Phule G 96022, 95123, 93208, 92028, BCP91 and 28 tolerant and the rest were found to be moderately susceptible and susceptible. Not much work has been done, so far, on the screening of chickpea germplasm against this pathogen except for several remark of Von Rheen *et al.* (1992) and Jayant Bhatt and Bhatt (1993) who reported ICC2644, 10384, 10630, 11224, 11332, ICCL 81002, 81010, ICC 1244, 12450, ICCV10 H355, BG M416, BG416, ICC1376 and ICC11314 were found to resistant. However, there is scope to use the genotypes found resistant in the present investigation, in resistance breeding programme against *R. bataticola* of chickpea.

In studies to find out the suitable fungitoxicant for the control of this disease, the efficacy of 21 fungitoxicants of various groups, including bio-agents, was evaluated first in bioassay test against *R. bataticola*. Out of which four namely, Carbendazim 25 DS, Carbendazim 25DS+ Thiram 80WP, Carboxin 75WP and Topsin-M 50WP proved to be the most effective as they inhibited the growth of the fungus completely. The rest of the fungitoxicants were categorized as partially effective amongst which Lasso gave the highest inhibition of the fungal growth and was significantly superior to the remaining ones. Thiram 80WP, Captan 50WP and Thiobendazole 60WP, in which Captan 50 WP and Thiobendazole 60 WP did not differ significantly from each other and Monocrotophas 40EC has was at par to Indosulfan in checking the fungal growth. The rest of the fungitoxicants in the descending order of efficacy were Phorate 10G, Chloropyriphos 20EC, *Bacillus subtilis*, Phosphomidan 85EC, Fungilin (*Trichoderma harzianum*), Ziram 80WP, Basalin, Isoproturan, Streptocycline and Blue copper-50. Streptocycline and Blue copper-50 were found to be statistically near about similar in their efficacy. The efficacy of various groups of fungitoxicants was evaluated against *R. bataticola* in bioassay test by several workers (Goyal and Mehrotra (1981), Sarwar and Raju (1985), and Kulkarni *et al.*, (1992) and these fungitoxicants inhibited the growth of the fungus to varying extents as observed in the present study.

The efficacy of eight seed dressing fungicides those found effective or partially effective in bioassay test, was evaluated under pot conditions to see their effect on seed germination and seedling infection by the pathogen. Although, all the seed dressing

fungicides improved seed germination to varying extents, but significant increase in seed germination was obtained in case of Carbendazim 25DS and Carbendazim 25 DS+Thiram 80WP treated seeds, followed by Carboxin 75WP, Thiram(1: 1) 80WP, Mancozeb 80WP and Captan50WP were also found effective in reducing the seedling infection and improving the seed germination significantly. However the remaining fungicides namely Topsin-M 50WP and Thiobendazole 60WP failed to control seedling infection of chickpea as they were at par with the untreated check, considering the results of both the years. The results are in close agreement with the findings of Taya *et al.*, (1990), who found that Carbendazim alone or in combination with Thiram assured treatment, pre-sowing, soil-drench and seed treatment + drench of chickpea gave the best control of *R. bataticola*. Peshney *et al.*, (1992) also reported that the *R. bataticola* was effectively eliminated by treated with Mancozeb, Carboxin, and Thiobendazole (all 0.2%). These results more or less support the present findings.

Thus on the basis of these results its is elicited that the dry root rot of chickpea can be effectively controlled by seed treatment with Carbonbendazim 25 DS or Carbendazim 25 DS+ Thiram 80WP.

SUMMARY

SUMMARY

Chapter-6

Chickpea (*Cicer arietinum* L.) is an important pulse crop of this country, diseases are one of the major factors responsible for reduction in yield of the chickpea crop. Different recommended high yielding variety as well as local varieties were observed to suffer from dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler.

Among the fungal diseases the incidence of dry root rot, has increased considerably on chickpea during recent years on which not much work has been done, so far. With a view to combating this serious malady for exploiting the full yield potential of the recommended cultivars of chickpea, the studies were taken up to find out the prevalence and incidence of the disease in various agro-climatic regions of Uttar Pradesh, to study morphological and cultural characters of the pathogen and its viability and virulence, to find out the optimum growth requirement of the pathogen, to ascertain the role of toxins in pathogenesis, to investigate the host range of the pathogen as well as to find out the sources of resistance and chemical control of the disease. The information obtained on various aspects in the present study is summarized here briefly.

The dry root rot of chickpea caused by *R. bataticola* (Taub.) Butler was observed in moderate to severe form in Uttar Pradesh as is evident from the surveys conducted during 1998-99 and 1999-2000 in different agro-climatic regions of this state. The survey revealed that the disease was wide spread in occurrence in U.P. with 19.00 to 42.00 per cent incidence. The highest disease incidence was recorded at Jaspura (Banda) and the lowest at the G.B. Pant Univ. of Agric. & Tech., Pant Nagar, Udham Singh Nagar, and Meeraganj (Bareilly). At other location the incidence varied between 19 and 42 per cent.

Under natural conditions, the disease appeared during late October to mid-November decreased in December and January, but it was again increased in February and March, when the temperature was increased. The symptoms of the disease were yellowing of the leaves, within a day or two such leaves dropped and in the course of another few days, plants were showed wilting symptom. The roots were when

examined closely dark, lesions seen. The plants were pulled out from the soil and examined, the basal stem and main root showed dry root rot symptom. The tissues were weakened and break-off easily. In advance stage sclerotial bodies scattered on internal parts of the affected tissues. The fungus was considered as a moderate to high temperature pathogen. The incidence of the disease increased with temperature between 28-35°C, the disease was appeared at high temperature. High temperatures coupled with low soil moisture increased in disease incidence double. The disease on chickpea plants were more severe in sandy soils than heavy soils.

The pathogen was readily isolated on Potato dextrose agar medium and its pathogenicity was proved following the Koch's postulates. It was observed that the incidence of the disease always increased with the increased age of the host.

The morphological and cultural character of the pathogen were studies on potato dextrose agar for identifications and it was found that light to dark gray coloured colonies with fluffy mycelium. Fast growing dark brown with ample sclerotial production margin smooth, covering full plate within 6-10 days. Mycelium was initially grayish white, became brown to dark brown with age, septated, profusely branched 4.5-7.5µm on Potato dextrose agar medium, septa were more abundant in older portion than the younger once. Branches arised often at right angle with constriction at their point of origin. Invariably there was presence of a septum close to the point of origin of the branches. Sclerotia were produced in abundance both in xylem of the roots and on Potato dextrose agar medium were dark brown, globose to oval or irregular in shape and dark in colour appeared somewhat hairy owing to emergence of dark hyphae from their surface measured 10.5-95.0µm in diameter in the xylem of the roots and was always smaller than the pycnidia produced in host tissues. Pycnidia were developed on host tissues in field, globose to oval, ostiolate, beaked or without beak light to dark brown in colour, superficial or submersed, scattered all over the surface of the roots, spores (conidia) were single celled, hyaline, oval, elongated to elliptical in shape, oozing out in the form of dull white cirrhous and measured 15.5 – 21 x 5.5 – 9.0 µm on the basis of morphological and cultural character, the fungus caused dry root rot of chickpea was identified as *Rhizoctonia bataticola* (Taub.) Butler.

Out of the eight culture media tested in solid form the best growth of the fungus was obtained on Potato dextrose agar medium, but it did not support any sporulation.

Good growth of the pathogen was obtained on Standard nutrient agar, Chickpea root extract agar medium, which did not significantly differ from each other. Fair growth of the pathogen was obtained on Asthana and Hawker's medium, whereas the remaining media viz., Carrot root extract agar, Oat meal agar, Czapek Dox agar and Richard's agar media support poor growth. No sporulation was observed on any of the solid media tested.

Of the eight culture media tested in liquid form the best growth of the fungus was obtained on Potato dextrose medium and was significantly superior to the rest of medium tested. Good growth of the pathogen was obtained on Standard nutrient and Chickpea root extract medium, which did not significantly differ from each other, fair growth of the pathogen was obtained on Asthana and Hawker's medium, whereas the remaining media viz; Carrot root extract, Oat meal, Czapeks Dox and Richard's media supported poor growth. No sporulation was observed on any liquid media tested.

In studies on the temperature requirement, it was noted that although the pathogen was able to grow over a wide temperature range, but the optimum for its growth was found to be 35°C. The minimum temperature at which the pathogen could grow was 15°C. Sporulation of the fungus was not observed at any temperature value.

Studies on pH requirement of the pathogen were also carried out and it was observed that, although, fungus could grow over a wide pH range varying from 3.0 to 9.0, but the optimum pH for its growth was found to be 5.5. The growth of the fungus was reduced both below and above the optimum pH value. However, no sporulation of the fungus occurred at any pH level from 3-9 tested.

Studies were taken up to find out the production of toxins by the pathogen *in vitro* and it was observed that this fungus was able to produce certain metabolites in Chickpea root extract medium, which when applied on host leaves produced necrotic spots within 24 hours after application.

The study on the effect of toxic filtrate on seed germination indicated that it inhibited seed germination and root and shoot length of the germinating seeds at various concentrations. The heated culture filtrate also caused inhibition of seed germination and root and shoot length. Although the inhibition effect was more

pronounced in case of unheated culture filtrate. This indicated that the inhibitory substance in the filtrate was only partially thermostable.

Studied on the mode of survival of the pathogen in nature and the role of seed, soil and diseased plant debris in primary infection of the disease were also carried out. It was observed that the pathogen remained viable in seed, soil and diseased plant debris, in infection stage for seven months till the next sowing season. Infected seeds, soil and plant debris were observed to serve as the source of primary inoculum, for disease development.

In the host range studies of *R. bataticola*, it was found that out of the 71 plant species belonging to 17 families tested the fungus was able to infect 46, but failed to infect 25 of them. It parasitized majority of the hosts of families viz; Papilionatae Cucurbitaceae, the pathogen could not infect the plants of the families Liliaceae and Linaceae. Thus it indicates that fungus has a very wide host range.

In order to locate the source of resistance against the dry root rot of chickpea 45 varieties/culture were first tested by Blotter paper technique in laboratory. Subsequently, the resistant and moderately resistant ones were further tested on condition of epiphytotics in the glasshouse (pots). Out of these, 20 viz; H355, JG315, Pusa212, G543, ICC32, BG244, Phule G5, IICC 2644, 10384, 11224, 1030, 11332, ICCV10, IVV12441, 12263, ICCL81010, 81002, Avrodhi, I LC 702 and 200 proved resistant five namely, GCP-9504, Phule G96313, 96105, 96020 and GL 91059 moderately resistance and the rest were found to be tolerant, moderately susceptible and susceptible.

In studies to find out the suitable fungitoxics for the control of this disease the efficacy of 21 fungitoxics of various groups tested against *R. bataticola*, was first tested *in vitro*, out of which four namely Carbendazim; 25DS, Carbendazim 25DS + Thiram 80WP, Carboxin 75WP and Topsin -M50WP and proved to be the most effective as they inhibited the growth of the fungus completely. Among the partially effective fungitoxics, Lasso gave the highest inhibition of the fungal growth and was significantly superior to the remaining ones. Thiram (1: 1) 80WP observed to be the next best, followed by Mancozeb 80WP, Captan 50WP and Thiobendazole 60WP, in which Captan 50WP and Thiobendazole 60WP did not differ significantly from each other and Monocrotophos 40EC was at par to Indosulfan 35EC in checking the fungal

growth. The rest of the fungitoxicants in the descending order of efficiency were Phorate 10G, Chloroprifos 20EC, *Bacillus subtilis*, Phosphomidan 85EC, Funginil (*Trichoderma harzianum*), Ziram 80WP, Basalin, Isoproturan, Streptocycline and Blue copper -50.

The efficacy of eight seed dressing fungicides those found effective or partially effective in bioassay test, was evaluated under pot condition to see their effect on seed germination and seedling infection by the pathogen. Although, all the seed dressing fungicides improved seed germination to varying extents, but significant increase in seed germination was obtained in case of Carbendazim 25DS and Carbendazim 25DS + Thiram 80WP treated seeds, followed by Carboxin 75WP, Thiram 80WP, Mancozeb 80WP and Captan 50WP were also found effective in reducing the seedling infection and improving the germination significantly. However, the remaining fungicides namely Topsin-M 50WP and Thiobendazole 60WP failed to control seedling infection of chickpea as they were at par with the untreated check considering the results of both the year.

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* Original not seen

APPENDICES

APPENDICES

Appendix 1: Composition of culture media.

Medium	Quantity of chemicals for 1000ml of the medium
(1).Potato dextrose:	
Peeled potato	200.00g
Dextrose	20.00g
Distilled water	1000.00ml
(2). Standard nutrient:	
Potassium hydrogen phosphate	1.36g
Sodium carbonate	5.00g
Magnesium sulphate	5.00g
Dextrose	5.00g
Asparagine	1.00g
Distilled water	1000.00ml
(3).Chickpea root extract:	
Chickpea root	200.00g
Distilled water	1000.00ml
(4). Carrot root extract	
Carrot root	200.00g
Distilled water	1000.00ml
(5).Oat meal:	
Oat meal	50.00g
Distilled water	1000.00ml
(6).Czapeks(Dox):	
Sodium nitrate	2.00g
Dipotassium hydrogen phosphate	1.00g
Potassium chloride	0.5g
Magnesium sulphate	0.5g
Ferrous sulphate	0.01g
Sucrose	30.00g
Distilled water	1000.00ml
(7) Richard's:	
Potassium nitrate	10.00g
Potassium dihydrogen phosphate	5.00g
Magnesium sulphate	2.50g
Ferric chloride	0.02g
Sucrose	50.00g
Distilled water	1000.00ml
(8). Asthana and Hawker's:	
Potassium nitrate	3.50g
Potassium dihydrogen phosphate	1.75g
Magnesium sulphate	0.75g
Glucose	5.00g
Distilled water	1000.00ml

Appendix- 2 ANOVA Tables for different experiments

Table-6. Solid media

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	7	6399.85	914.26	643.84***
Error	16	31.71	1.98	
Total	23	6431.56		

Table-7. Liquid media

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	7	267.47	38.21	113.04***
Error	16	5.42	0.338	
Total	23	272.89		

Table-8. Temperature levels

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	9	717.04	79.67	69.15***
Error	20	24.45	1.22	
Total	29	741.49		

Table-9. pH values

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	11	1072.92	97.54	187.57***
Error	24	12.38	0.52	
Total	35	1085.30		

Table-20. Bioassay test

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	21	41445.72	1973.60	2517.062***
Error	44	34.50	0.7840	
Total	65	41480.22		

Table-21. Seed germination(1998-99)

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	8	615.15	76.98	6.72***
Error	27	308.86	11.44	
Total	35	924.01		

Table-21. Seed germination(1999-2000)

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	8	680.79	85.09	5.59***
Error	27	410.68	15.21	
Total	35	1091.47		

Table-21. Seedling infection (1998-99)

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	8	2104.58	263.07	7.00***
Error	27	1013.82	37.54	
Total	35	3118.40		

Table-21. Seedling infection (1999-2000)

Source of variation	D.F.	S.S.	M.S.S.	F
Treatment	8	1833.36	229.17	6.71***
Error	27	921.54	34.13	
Total	35	2754.90		

Appendix 3: Fungitoxicants and their ingredients

Groups	Fungitoxicants	
(A) Dithiocarbamate compounds	Fungicides Thiram (1:1) 80 WP Mancozeb 80WP	75% Tetramethy thiuram disulphide. 80% Zinc and Maganese ethylene bis dithiocarbamate.
Copper compounds Systemic	Ziram 80WP Blue coper-50 Carbendazim 25DS Topsin-M 50WP	80% Zinc dimethyle dithiocarbamate. 50% Copper as copper oxychloride. Methyl-2 benzimidazole carbamate. 1-2-bis (3-methoxy carbony 1-2-thioureidobenzene.
Antibiotic Heterocyclic Nitrogen compounds (B)	Thiobendazole 60WP Carboxin75 WP (Vitavax) Carbendazim 50WP + Thiram(1:1)80WP. Streptocycline (Streptomycin) Captan 50WP.	2-(4-thiozoly)-benzimidazol (TBZ). 5,6-dihydro-2-methyle-1,4-oxathin 3-carboxanilide Methyle-2 benzimidazole carbamate + 75% Tetramethyl thiuram disulphide. Produced by <i>Streptomyces giseus</i> 75% N-(trichloro methyl thio)- 4 cyclohexane-1,2-dicarboximide.
	Non-target pesticides 1. Insecticides Monocrotophas 40EC. Indosulfan-35EC. Phorate 10G. Chloropyrifos 20EC. Phosphamidan 85EC.	Dimethyle cis-1-methyle-2 methylecarbamoyninyl phosphate. 6,7,8,9,10,10-hexachloro-1,5,5a,6,9,9a- hexahydro-6,9 methano-2,4,3-benzo (e)- dioxathepin-3 oxide. Diethyl S- (ethyl- thiomethyle) phosphorothiolothiolothionate. O,O- diethyle O-(3,5,6- trichloro- 1- pyridyl) phosphorothioate O-(2-chloro-2diethyl-carbamoyl) -1- methyl-vinyl) -O,O- dimethyl phosphate.
(C)	2 Herbicides Lasso Basalin Isoproturan Bio-agents <i>Bacillus subtiles</i> Funginil (<i>Trichoderma</i> sp.)	2, chloro-2,6- diethyl -N-(Methoxy methyl) acetanilide. N- (2- chloroethyl) -2,6- dinitro-N- propyl -4 (trifluoro methyl aniline). 4- iso propyl phenyl Division- Firmucules, Class- Firmibacteria, simple gram- positive. Sub-division- Deuteromycotina, Class- Hyphomycetes, Order- Hyphales (Moniliales), Family- Moniliaceae,